
Design of Experiments and Quality Assurance

IE 1072, Fall 2025

University of Pittsburgh

Instructor: Prof. Amin Rahimian | Scribe: Donghwan Kim

Course Purpose

These notes accompany IE 1072 at the University of Pittsburgh and are organized around the technical aspects of two complementary subjects: the design of experiments (DOE) and statistical quality assurance. The course provides IE majors with a working background in designed experiments and the methods used to monitor and control quality in production processes, while deliberately leaving the broader managerial and process-improvement aspects of quality assurance to companion courses (IE 1102 Lean Six Sigma and IE 1076 Total Quality Management).

Throughout these notes, probability and statistics are treated as tools rather than ends in themselves: each probabilistic result is introduced because it underwrites a downstream decision about whether a factor effect is real, whether a lot meets its specification, or whether a process has drifted out of control. The text moves from the probability and inference machinery needed to make those decisions (Parts I–VI) to the design of experiments that produce defensible data (Part VII), and finally to the lot-acceptance and process-control procedures used to act on that data in industrial practice (Parts VIII–IX). R is used throughout as a working notebook for the same calculations. Companion scripts are linked inline where the corresponding technique first appears.

Learning Objectives

By the end of the course, students should be able to:

1. Describe the purpose, importance, and benefits of design of experiments (DOE), statistical process control (SPC), and acceptance sampling.
2. Define key terms associated with DOE, SPC, and acceptance sampling.
3. Explain and conduct a well-designed statistical experiment.
4. Articulate the seven phases of the DOE framework.
5. Understand the role that assumptions play in the accuracy, precision, and validity of results related to DOE, SPC, and acceptance sampling.

Contents

Part I

Probability Foundations

Week 1 · What We Cover in This Part

This part develops the probabilistic machinery that the rest of the course relies on. We start with discrete and continuous random variables, move on to the mean and variance and their behavior under linear operations, and finish with the four distributions that recur throughout DOE, SPC, and acceptance sampling: the binomial, Poisson, hypergeometric, and normal.

Textbook: Montgomery, Ch. 3, pp. 75–93.

1 Random Variables

A *random variable* maps the outcome of a random experiment to a number on the real line or to an integer.

1.1 Discrete (Arithmetic) Random Variables

A discrete random variable Y takes integer values and is described by its probability mass function (PMF):

$$P(Y = i) = p_i, \quad i = 0, 1, 2, \dots, \quad \sum_{i=0}^{\infty} p_i = 1.$$

1.2 Continuous Random Variables

A continuous random variable X takes values on the real line and is described by its probability density function (PDF) f :

$$P(x_1 < X \leq x_2) = \int_{x_1}^{x_2} f(x) dx, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

1.3 Cumulative Distribution Function

The cumulative distribution function (CDF) is

$$F(x) = P(X \leq x).$$

F is non-negative, monotone non-decreasing, and bounded above by 1. It assigns probabilities to half-closed intervals:

$$P(x_1 < X \leq x_2) = F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(x) dx.$$

For a discrete random variable,

$$F(x) = \sum_{i \leq x} p_i.$$

2 Mean and Variance

2.1 Definitions

The *mean* (expected value) of a random variable X is

$$E(X) = \begin{cases} \int_{-\infty}^{\infty} x f(x) dx & \text{(continuous),} \\ \sum_{i=0}^{\infty} i p_i & \text{(discrete).} \end{cases}$$

The *variance* measures dispersion around the mean:

$$\text{Var}(X) = E[(X - E(X))^2] = E(X^2) - (E(X))^2.$$

2.2 Bernoulli Random Variable

Let $X = 1$ with probability p and $X = 0$ with probability $1 - p$. Its mean and variance summarize the building block from which the binomial, geometric, and negative-binomial distributions are constructed.

Bernoulli Moments

$$E(X) = p, \quad \text{Var}(X) = p(1 - p) \leq \frac{1}{4}.$$

The variance is maximized at $p = 1/2$, where the outcome is most uncertain.

2.3 Linear Operations

For any constant a ,

$$E(aX) = a E(X), \quad \text{Var}(aX) = a^2 \text{Var}(X).$$

Proof of $\text{Var}(aX) = a^2 \text{Var}(X)$. With $\mu = E(X)$, $E(aX) = a\mu$, so

$$\text{Var}(aX) = E[(aX - a\mu)^2] = E[a^2(X - \mu)^2] = a^2 \text{Var}(X). \quad \square$$

2.4 Variance of a Sum

For random variables $X_1, \dots, X_n$,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

If the X_i are pairwise uncorrelated, i.e. $\text{Cov}(X_i, X_j) = E[(X_i - EX_i)(X_j - EX_j)] = 0$ for all $i \neq j$, then

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Independence implies pairwise uncorrelatedness. The converse is not generally true.

3 Important Distributions

3.1 Binomial Distribution

A *binomial* random variable counts the number of successes in n independent and identically distributed (i.i.d.) Bernoulli(p) trials:

$$S_n = \sum_{i=1}^n X_i, \quad X_i \sim \text{Bernoulli}(p).$$

Binomial Distribution

$S_n \sim \text{Binomial}(n, p)$ has PMF, mean, and variance

$$P(S_n = i) = \binom{n}{i} p^i (1-p)^{n-i}, \quad E(S_n) = np, \quad \text{Var}(S_n) = np(1-p).$$

R companion: [01-binomial-pmf-cdf.R](#)¹

```
pmf <- dbinom(x, size = 10, prob = 0.25)    # PMF at x = 0..10
cdf <- pbinom(x, size = 10, prob = 0.25)    # CDF
random_samples <- rbinom(n = 100, size = 10, prob = 0.25)
median(random_samples); mean(random_samples)
```

3.2 Poisson Distribution and the Sum of Independent Poissons

If $X_1, X_2, \dots, X_n$ are independent with $X_i \sim \text{Poisson}(\lambda)$, then their sum is again Poisson:

$$\sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda).$$

This stability under addition makes the Poisson convenient for modeling counts aggregated over independent units (e.g. accident counts across drivers, defect counts across batches). To assess whether an observed total is consistent with a hypothesized common rate λ , one compares the observed sum to the $\text{Poisson}(n\lambda)$ tail probabilities.

R companion: [02-poisson-accidents.R](#)²

```
avg <- sum(num_accidents * people_accident_counts) / sum(people_accident_counts)
poisson_probs <- dpois(num_accidents, lambda = avg)
plot(total_accident_count * poisson_probs, people_accident_counts) # expected vs observed
```

3.3 Hypergeometric Distribution

The hypergeometric distribution arises in finite-population sampling *without replacement*: in a population of size N there are K marked items, and a random sample of size n is drawn.

¹<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/01-binomial-pmf-cdf.R>

²<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/02-poisson-accidents.R>

Hypergeometric Distribution

$X \sim H(N, K, n)$ counts marked items in a sample of size n drawn without replacement from a population of size N containing K marked items:

$$P(X = i) = \frac{\binom{K}{i} \binom{N-K}{n-i}}{\binom{N}{n}}, \quad E(X) = \frac{nK}{N}.$$

When $N \gg n$, $H(N, K, n) \approx \text{Binomial}(n, K/N)$.

Estimating the quality of a lot. If N and n are known and i is observed, the unknown is the total number of defective items K .

Capture-recapture. If K animals are tagged and released and i are recaptured in a sample of size n , the unknown is the total population size N .

3.4 Normal Distribution

A standard normal random variable Z has PDF

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

We write $Z \sim N(0, 1)$, and $X = \mu + \sigma Z \sim N(\mu, \sigma^2)$.

R companion: [03-normal-distribution.R](#)³

```
low <- pnorm(17, mean = mean(data$Torque), sd = sd(data$Torque))
up   <- pnorm(19, mean = mean(data$Torque), sd = sd(data$Torque))
up - low    # P(17 < X <= 19) under the fitted normal
```

The distributions and moments collected here are the building blocks. Real data, however, comes in finite samples. Part II asks how to extract reliable information from such samples: how the sample mean $\bar{X}$ relates to the population mean μ , why representative sampling can outperform simple random sampling, and how the maximum likelihood principle picks the parameter value that best explains what we have observed.

³<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/03-normal-distribution.R>

Part II

Sampling, Estimation, and Maximum Likelihood

Weeks 1–2 · What We Cover in This Part

This part connects sample statistics to the population parameters they estimate, and asks where their uncertainty comes from. We define the sample mean and sample variance and show that S^2 is unbiased. We then contrast within- and between-stratum variance to explain why representative (stratified) sampling can outperform simple random sampling, and apply the maximum likelihood principle to the Bernoulli/Binomial and hypergeometric parameters. Two motivating problems thread through these sections: Fisher's *Lady Tasting Tea* and Benford's leading-digit law.

Textbook: Montgomery, Ch. 3, pp. 100–102, and Ch. 4, pp. 104, 109.

4 Sample versus Population

	Sample	Population
Size	n observations $X_1, \dots, X_n$	
Mean	$\bar{X} = \frac{1}{n} \sum_i X_i$	μ
Variance	$S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$	σ^2

When the X_i are i.i.d. with mean μ and variance σ^2 , the sample mean is unbiased and its variance shrinks like $1/n$.

Mean and Variance of $\bar{X}$

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

The standard deviation of $\bar{X}$ is therefore $\sigma/\sqrt{n}$, which is the $1/\sqrt{n}$ scaling that drives the LLN, the CLT, and every CI we will see.

Proof. By independence,

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}. \quad \square$$

Remark 4.1 (Bias does not vanish with more samples). $\bar{X}$ is unbiased, so its expected error around μ tends to zero as $n \rightarrow \infty$. By contrast, a *biased* estimator retains its bias regardless of how many samples are collected: increasing n only reduces the variance, not the systematic offset.

4.1 Unbiasedness of S^2

The denominator $n-1$ in the sample variance is not a typo: it is the correction needed to make S^2 an unbiased estimator of σ^2 when μ has been replaced by its estimate $\bar{X}$.

Unbiasedness of S^2

For i.i.d. samples $X_1, \dots, X_n$ with variance σ^2 ,

$$E(S^2) = \sigma^2, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The dividing constant $n-1$ is the *degrees of freedom*: one degree is consumed by estimating $\bar{X}$.

Proof. Write $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$. Then

$$E[(X_i - \bar{X})^2] = E[(X_i - \mu)^2] + E[(\bar{X} - \mu)^2] - 2E[(X_i - \mu)(\bar{X} - \mu)] = \sigma^2 + \frac{\sigma^2}{n} - 2 \cdot \frac{\sigma^2}{n} = \sigma^2 \left(1 - \frac{1}{n}\right),$$

using $E[(X_i - \mu)(\bar{X} - \mu)] = \text{Cov}(X_i, \bar{X}) = \sigma^2/n$. Summing over i and dividing by $n-1$,

$$E(S^2) = \frac{1}{n-1} \cdot n \cdot \sigma^2 \left(1 - \frac{1}{n}\right) = \sigma^2. \quad \square$$

5 Within/Between Variance and Representative Sampling**Story: Opportunity Samples in Bombay and Washington**

Why is random sampling preferred to “representative” sampling by judgment? Karl Pearson’s early biometric programme had relied on what is now called an *opportunity sample*: *Biometrika* correspondents would collect whatever skulls happened to be available in a graveyard, pour shot into them to measure cranial capacity, and send the numbers in. “The data were those that were most easily available. They did not have to be truly representative of the entire distribution.”

A specific failure of opportunity sampling surfaced in early-1930s India. Bales of jute were assembled on the Bombay docks for shipment to Europe, and each bale was valued by sampling its interior with a hollow blade. But in shipping the outsides deteriorated and the cores compacted; the blade tended to be deflected into the outer, damaged region. “The opportunity sample was biased in favor of finding inferior jute when the quality of the bale was much higher.” Prasanta Chandra Mahalanobis, who had encountered the problem on the railroad shipping jute to the docks, used it as a standard illustration of why opportunity samples cannot be trusted.

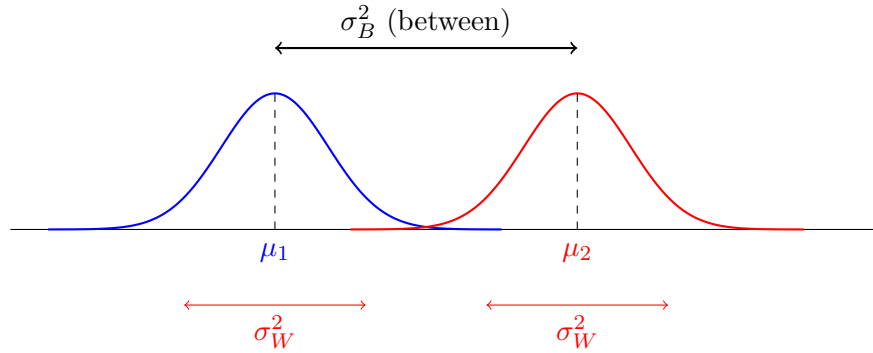
Mahalanobis’s answer, developed at the Indian Statistical Institute (which he founded in 1931 on the grounds of one of his family estates), was the *random* sample: pick individuals out of the larger population by a randomizing mechanism, then exploit the known probability distribution of the random sample to compute confidence bounds on the truth. “The random sample is better than either the opportunity sample or the judgment sample, not because it guarantees correct answers but because we can calculate a range of answers that will contain the correct answer with high probability.”

The same lesson was learned in the United States during the New Deal. Young statisticians at the Bureau of Labor Statistics, drawing on Neyman’s papers, designed a survey of 2% of the postal routes for the 1937 Unemployment Census, and showed that this small random sample was more accurate than the judgment samples it replaced. The Census Bureau and the

Bureau of Labor Statistics built the modern apparatus of sample surveys on that foundation.

Source: Salsburg (2001), The Lady Tasting Tea, ch. 17, pp. 169–175.

Suppose a population is partitioned into two strata with means μ_1, μ_2 and a common within-stratum variance σ_W^2 . Let stratum k have proportion P_k in the population, with $P_1 + P_2 = 1$.



Each observation X_i in a simple random sample admits the decomposition

$$X_i = Y_i + Z_i, \quad Y_i = \begin{cases} \mu_1 & (P_1) \\ \mu_2 & (P_2) \end{cases}, \quad E(Z_i) = 0, \quad \text{Var}(Z_i) = \sigma_W^2.$$

A direct computation yields

$$\text{Var}(Y_i) = P_1 P_2 (\mu_1 - \mu_2)^2 =: \sigma_B^2, \quad \text{Var}(X_i) = \sigma_W^2 + \sigma_B^2.$$

Stratified (representative) sampling. If the sample is forced to contain $n_k = nP_k$ observations from stratum k , the stratified sample mean is

$$\bar{X}_{\text{rep}} = \frac{n_1}{n} \bar{X}_1 + \frac{n_2}{n} \bar{X}_2,$$

with variance

$$\text{Var}(\bar{X}_{\text{rep}}) = \left(\frac{n_1}{n}\right)^2 \frac{\sigma_W^2}{n_1} + \left(\frac{n_2}{n}\right)^2 \frac{\sigma_W^2}{n_2} = \frac{\sigma_W^2}{n}.$$

Variance Reduction from Stratified Sampling

Under simple random sampling, $\text{Var}(\bar{X}) = (\sigma_W^2 + \sigma_B^2)/n$. Under stratified sampling with $n_k = nP_k$,

$$\text{Var}(\bar{X}_{\text{rep}}) = \frac{\sigma_W^2}{n}.$$

The between-stratum variance σ_B^2 is eliminated: forcing the sample composition to match the population's removes the part of the variance attributable to stratum membership. This is why representative sampling beats simple random sampling whenever strata differ in mean.

R companion: [04-representative-sampling.R](#)⁴

⁴<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/04-representative-sampling.R>

```

for (i in 1:n) {
  # random sampling
  if (runif(1) < p_1) random_sampling[i] <- rnorm(1, mu_1, sd1)
  else
    random_sampling[i] <- rnorm(1, mu_2, sd2)
}
# representative sampling: replace `runif(1) < p_1` with `i <= n * p_1`.
var(random_sampling); var(represenatative_sampling)

```

6 The Likelihood Function and Maximum Likelihood

Definition 6.1. The *likelihood function* is the probability of the observed data viewed as a function of the unknown parameter:

$$L(\theta) = P(\text{data} \mid \theta).$$

The *maximum likelihood estimator* (MLE) is

$$\hat{\theta} = \arg \max_{\theta} L(\theta).$$

6.1 MLE for Bernoulli/Binomial

If $S_n \sim \text{Binomial}(n, p)$ and we observe $S_n = k$, then

$$L(p) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Differentiating $\log L$ in p and setting to zero:

$$\frac{k}{p^*} - \frac{n-k}{1-p^*} = 0 \implies \hat{p} = \frac{k}{n}.$$

MLE for the Binomial Probability

With $S_n \sim \text{Binomial}(n, p)$ and observed k successes,

$$\hat{p} = \frac{k}{n}.$$

The MLE is the empirical success rate, and it is unbiased: $E(\hat{p}) = p$.

6.2 MLE for the Hypergeometric Parameter K

For $X \sim H(N, K, n)$ with N, n known and $X = i$ observed,

$$h_K := P(X = i \mid K) = \frac{\binom{K}{i} \binom{N-K}{n-i}}{\binom{N}{n}}.$$

Comparing successive values,

$$\frac{h_{K+1}}{h_K} > 1 \iff K < \frac{i(N+1)}{n} - 1,$$

so h_K is increasing then decreasing in K , and the integer-valued maximizer is

$$\hat{K} = \left\lfloor \frac{i(N+1)}{n} \right\rfloor.$$

For large N with $K/N \rightarrow p$, the hypergeometric tends to $\text{Binomial}(n, p)$ and $\hat{p} = i/n$, recovering the Bernoulli MLE.

MLE for the Hypergeometric Parameter K and Capture–Recapture

For $X \sim H(N, K, n)$ with N, n known and $X = i$ observed,

$$\hat{K} = \left\lfloor \frac{i(N+1)}{n} \right\rfloor.$$

For estimating an unknown population size N given K tagged animals, a recapture sample of size n , and i tagged animals observed in the recapture, the MLE is

$$\hat{N} = \left\lfloor \frac{K n}{i} \right\rfloor.$$

Practice 1: Estimating a Trout Population

A wildlife agency tags K trout, releases them, and a short time later captures a recapture sample of size n from the same waters, of which i are tagged. Suggest a principled estimator for the total trout population N , justify your choice using the hypergeometric likelihood, and report the resulting estimate as a function of K , n , i .

7 Lady Tasting Tea

Story: Fisher at Rothamsted, and the Lady Tasting Tea

A summer afternoon in Cambridge, England, in the late 1920s. A group of university dons, their wives, and guests are sitting around an outdoor table for afternoon tea. One of the women insists that tea tastes different depending on whether the tea is poured into the milk or the milk into the tea. The scientific minds at the table scoff. A man at the table, “thin, short . . . with thick glasses and a Vandyke beard beginning to turn gray,” pounces on the question and says, “Let us test the proposition.”

The man was Ronald Aylmer Fisher, then in his late thirties. The story is preserved because one of the guests, H. Fairfield Smith, later a professor of statistics at the University of Connecticut, recounted it to David Salsburg in the 1960s.

At the time Fisher was working at the Rothamsted Agricultural Experimental Station, which had been running fertilizer trials (“artificial manures” in the period term) for almost ninety years. Each station had its own “fertility index” for adjusting yields across seasons, and partisans of each index disputed which one was best. In a 1921 paper in the *Annals of Applied Biology*, Fisher showed that the competing indexes were algebraically the same correction, and that none of them was adequate. He then showed that year-to-year weather variation

swamped year-to-year fertilizer variation: the two effects were, to use the word Fisher himself coined, *confounded*. “Ninety years of experimentation and over twenty years of scientific dispute had been an almost useless waste of effort.”

Fisher's response was *The Design of Experiments* (1935), whose second chapter works through how to test the lady's claim. The mathematics of that test is what this section develops. And the lady? Fisher's book does not say. But Smith told Salsburg that, on that summer afternoon, “the lady identified every single one of the cups correctly.”

Source: Salsburg (2001), The Lady Tasting Tea, ch. 1, pp. 1–8.

A taster is presented with 8 cups of tea, 4 with milk added first and 4 with tea added first, in random order. Let X be the number of milk-first cups identified correctly out of the 4 chosen. Under the null that the taster has no discriminating ability, $X \sim H(8, 4, 4)$:

$$P(X = 4) = \frac{\binom{4}{4}\binom{4}{0}}{\binom{8}{4}} = \frac{1}{70} \approx 0.014, \quad P(X \geq 3) = \frac{1}{70} + \frac{16}{70} = \frac{17}{70} \approx 0.243.$$

At the standard 5% significance level, only the perfect outcome $X = 4$ rejects the null. Observing 3 correct identifications does not constitute statistically significant evidence of discrimination.

8 Benford's Leading Digit Law

For a number $X \geq 1$, write $X = 10^{n+f}$ with $n = \lfloor \log_{10} X \rfloor$ and $f = \log_{10} X - n \in [0, 1)$. The leading digit equals K exactly when

$$\log_{10} K \leq f < \log_{10}(K + 1).$$

For data spanning many orders of magnitude in a way that makes f approximately uniform on $[0, 1)$, the proportion of values whose leading digit is K is approximately $\log_{10}(1 + 1/K)$. This empirical regularity (*Benford's law*) provides a fingerprint of naturally occurring data sets and is used in forensic accounting to flag fabricated numbers.

Practice 2: Benford on Real Data

Take a public dataset of counts that span several orders of magnitude (daily case counts of a disease, county-level vote totals, river-discharge volumes, accounting line items). Extract the leading digit of each entry, plot the empirical histogram of leading digits, and compare it against the Benford prediction $P(d) = \log_{10}(1 + 1/d)$ for $d = 1, \dots, 9$. Discuss any deviation: is it consistent with sampling noise, or does it warrant a closer look?

R companion: [05-benford-leading-digits.R](#)⁵

```
# Extract the leading digit of each county-level vote total
votes_dem_bnfdr <- extract.digits(votes$votes_dem, number.of.digits = 1)
```

⁵<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/05-benford-leading-digits.R>

```
# Theoretical Benford probabilities  $P(d) = \log_{10}(1 + 1/d)$   
for (i in 1:9) print(log10(1 + 1 / i))
```

We have seen the MLE pick a single best estimate. But how close is that estimate, in fact, to the truth, and how much does it improve as we collect more samples? Part III answers both questions through Chebyshev's inequality (which gives a crude bound that requires no distributional assumption), the Law of Large Numbers (which guarantees convergence), and the Central Limit Theorem (which gives the right rate, $1/\sqrt{n}$, with a normal limit shape).

Part III

Limit Theorems and Concentration

Weeks 2–3 · What We Cover in This Part

This part studies how sample averages behave as n grows. We cover Chebyshev's inequality (a distribution-free bound), the triangle of polling, the Law of Large Numbers (*whether* $\bar{X}$ converges), and the Central Limit Theorem (the *shape* and the $1/\sqrt{n}$ rate of the residual error). We close with the normal approximation to the binomial and the Galton board.

Textbook: Montgomery, Ch. 3, pp. 100–102.

Week 3 introduces R as a working notebook. The first two companions below collect the R/dplyr/ggplot2 idioms (indexing, the pipe, `filter/select/mutate/summarise`, histograms, `t.test`, `pnorm/qnorm`) and the time-series data-manipulation patterns (`scale_x_date`, date filtering, `mutate+ifelse`) that recur throughout the course.

R companion: [00-r-basics-tutorial.R](#)⁶

```
library(dplyr); library(ggplot2)
data %>% filter(Petal_width >= 0.4)           # row filter
data %>% mutate(ratio = Petal_width / Petal_length)
data %>% summarise(mean = mean(Petal_width, na.rm = TRUE))
ggplot(data, aes(x = Petal_width)) + geom_histogram(binwidth = 0.1)
```

R companion: [06-time-series-data-manipulation.R](#)⁷

```
timestamp_data <- timestamp_data %>%
  mutate(busy_status = ifelse(cases > 50, "high", "low"))
ggplot(timestamp_data, aes(x = date, y = cases)) +
  geom_line() + scale_x_date()
timestamp_data %>% filter(date >= as.Date("2023-07-01"))
```

9 Chebyshev's Inequality

Chebyshev's Inequality

Let X be a random variable with finite mean μ and variance σ^2 . For any $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Equivalently, with $\varepsilon = k\sigma$, $P(|X - \mu|/\sigma \geq k) \leq 1/k^2$.

⁶<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/00-r-basics-tutorial.R>

⁷<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/06-time-series-data-manipulation.R>

Proof (continuous case).

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \geq \int_{|x - \mu| \geq \varepsilon} (x - \mu)^2 f(x) dx \geq \varepsilon^2 \int_{|x - \mu| \geq \varepsilon} f(x) dx = \varepsilon^2 P(|X - \mu| \geq \varepsilon). \quad \square$$

The bound makes no distributional assumption beyond a finite variance, and so is necessarily quite loose for nice distributions like the normal.

10 Triangle of Polling

Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p) and $S_n = \sum_i X_i$. We seek conditions on n such that S_n/n is within ε of p with probability at least $1 - \delta$:

$$P\left(\left|\frac{S_n}{n} - p\right| \geq \varepsilon\right) \leq \delta.$$

10.1 Chebyshev-Based Bound

Since $\text{Var}(S_n/n) = p(1-p)/n$, Chebyshev gives

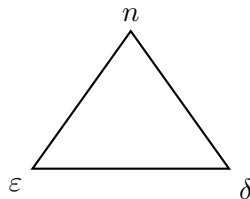
$$P\left(\left|\frac{S_n}{n} - p\right| \geq \varepsilon\right) \leq \frac{p(1-p)}{\varepsilon^2 n} \leq \frac{1}{4\varepsilon^2 n}$$

(using $p(1-p) \leq 1/4$). Setting the right-hand side equal to δ and solving for n :

Sample Size (Chebyshev Triangle of Polling)

$$n^* = \frac{1}{4\varepsilon^2 \delta}.$$

The three corners n , ε , δ form a triangle: any two determine the third. The required sample size does *not* depend on the population size N .



The polling paradox. The required sample size n^* does *not* depend on the population size N . This is striking, but in tight elections (where $p \approx 1/2$ and the required accuracy ε is tiny), the formula demands a sample size that is logistically infeasible. For $\varepsilon \sim 10^{-4}$ and $\delta \sim 10^{-2}$, one would need $n \sim 10^{10}$.

11 The Law of Large Numbers

Weak Law of Large Numbers

For i.i.d. $X_1, X_2, \dots$ with mean μ and finite variance σ^2 , and for any $\varepsilon > 0$,

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

The strong law states that $\bar{X} \rightarrow \mu$ almost surely, but its proof is beyond the scope of these notes.

12 The Central Limit Theorem

Story: Proving the Bell-Shaped Curve

The central limit theorem is the assertion that the sample mean, suitably standardized, approaches the normal distribution regardless of where the original data came from. Abraham de Moivre proved it for the binomial in the late eighteenth century. “For the next 150 years, no further progress was made on proving this conjecture.”

Fisher’s likelihood theory and Laplace’s least-squares justification both assumed the theorem. So did most of statistics. In the 1920s and 1930s a group of mathematicians in Scandinavia, Germany, France, and the Soviet Union (William Feller, Richard von Mises, Andrei Kolmogorov, Paul Lévy, Jarl Waldemar Lindeberg, Abraham Wald, Guido Castelnuovo, and others) attacked the conjecture in correspondence and at meetings. Lindeberg and Lévy independently arrived at a set of sufficient conditions, now called the Lindeberg–Lévy conditions, and by 1934 “the central limit theorem(s) was no longer a conjecture.” There was not a single theorem but a group, each derived from a slightly different set of conditions.

It is still hard work to check the Lindeberg–Lévy conditions for a specific statistic. The decisive simplification came in 1948, when Wassily Hoeffding published “A Class of Statistics with Asymptotically Normal Distribution” in the *Annals of Mathematical Statistics*. Hoeffding defined a class of statistics (now called U -statistics) and showed that any statistic in the class automatically satisfies the Lindeberg–Lévy conditions. “Since the publication of Hoeffding’s paper, almost all articles that show that a new statistic has a normal distribution do so by showing that the new statistic is a U -statistic.”

The practical upshot for this section: when these notes invoke the CLT to justify a normal approximation, they rely on results that were not fully proved until the 1930s, and on a tool kit (Hoeffding’s U -statistics, and later rate bounds) developed across the postwar decades. The bell-shaped curve is older than its proof.

Source: Salsburg (2001), The Lady Tasting Tea, ch. 9, pp. 83–89.

Central Limit Theorem

For i.i.d. $X_1, \dots, X_n$ with mean μ and finite variance σ^2 ,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

The deviation $|\bar{X} - \mu|$ is on the scale $\sigma/\sqrt{n}$, with a normal limit shape regardless of the

underlying distribution of the X_i .

12.1 CLT-Based Bound for Polling

Applying the CLT to S_n ,

$$\frac{S_n/n - p}{\sqrt{p(1-p)/n}} \xrightarrow{d} N(0, 1).$$

For $\alpha = 0.05$, $z_{0.975} = 1.96$, and

$$P\left(p \in \left(\frac{S_n}{n} - 1.96\sqrt{\frac{p(1-p)}{n}}, \frac{S_n}{n} + 1.96\sqrt{\frac{p(1-p)}{n}}\right)\right) \approx 0.95.$$

For a 95% confidence interval of half-width ε ,

$$\frac{1.96}{2\sqrt{n}} < \varepsilon \iff n > \frac{(1.96)^2}{(2\varepsilon)^2} \approx \frac{1}{\varepsilon^2}.$$

Sample Size (CLT Triangle of Polling)

$$n^* \approx \frac{1}{\varepsilon^2} \quad (\text{at } 95\% \text{ confidence}).$$

This is roughly $5\times$ smaller than the Chebyshev bound. The normal approximation extracts more information per observation, at the cost of an asymptotic (rather than universal) guarantee.

Practice 3: Sample Size for a Wildlife Survey

A wildlife agency wants to estimate the fraction of fish in a region that are of a certain species. They will conduct a phone survey of licensed anglers and tally the species of each fish reported. They will report their estimate with an absolute error of no more than 5% at confidence level at least 95%.

- Determine the required sample size using the inequality of Chebyshev.
- Determine the required sample size using a normal approximation (CLT).
- Suppose the true fraction is in fact $p = 0.55$. Compute the probability that the sample fraction lies within 5% of the truth for the two values of n in (a) and (b), and discuss which one is more wasteful.

Justify all your answers mathematically.

12.2 Distinction: LLN versus CLT

- LLN:** $\bar{X} \rightarrow \mu$ in probability (or almost surely). It tells us *whether* $\bar{X}$ converges, not how the residual error is distributed.

- **CLT:** $\sqrt{n}(\bar{X} - \mu)/\sigma$ has a standard normal limit. It describes the *shape* and *scale* of the residual error.

R companion: `07-normal-approx-binomial-clt.R`⁸

```
# Each binomial draw is itself a sum of binom_size i.i.d. Bernoullis.
binomial_samples <- rbinom(n = 100, size = 100, prob = 0.55)
hist(binomial_samples)
qqnorm(binomial_samples); qqline(binomial_samples)
```

13 Random Walks and the Galton Board

Let $X_i = \pm 1$ each with probability $1/2$, and $S_n = \sum_{i=1}^n X_i$. Then $E(X_i) = 0$ and $\text{Var}(X_i) = 1$, so $E(S_n) = 0$ and $\text{Var}(S_n) = n$. By the CLT,

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, 1).$$

The transformation $Y_i = (X_i + 1)/2 \in \{0, 1\}$ shows that

$$\frac{S_n + n}{2} \sim \text{Binomial}(n, 1/2),$$

so the random walk is a re-centering of a fair-coin Binomial. In a Galton board with n levels and N balls, the number of balls landing in the central slot is approximately

$$N \cdot \left(2\Phi\left(\frac{1}{2\sqrt{n}}\right) - 1 \right).$$

The CLT tells us that $\bar{X}$ deviates from μ on the scale $\sigma/\sqrt{n}$, and that the rescaled deviation is asymptotically normal. Part IV converts that scaling into a guaranteed-coverage interval: a range of values for μ that contains the truth with prescribed probability. The route runs through pivots, the χ^2 distribution, and the t distribution that “Student” invented for small-sample work at Guinness.

⁸<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/07-normal-approx-binomial-clt.R>

Part IV

Confidence Intervals

Weeks 4–5 · What We Cover in This Part

This part develops the active machinery for quantifying uncertainty about a parameter from data. We introduce pivotal quantities and use them to build confidence intervals for the mean and variance of a normal population in every combination of known and unknown nuisance parameters. Along the way we meet the t -distribution, handle differences in two normal means (known, common, and unequal variances) and in two proportions, and use the F -distribution for the ratio of two variances. We close with the structural relationship between the T , χ^2 , and F distributions.

Textbook: Montgomery, Ch. 4, pp. 111–128.

14 Pivotal Quantities

A *pivotal quantity* is a function of the data and the unknown parameter whose distribution does not depend on the parameter. Pivots are the engine of confidence interval construction: from a known reference distribution, one inverts the probability statement to bracket the parameter.

15 Mean of a Normal Population, Known Variance

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with σ^2 known. Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

is a pivot. For coverage $1 - \alpha$, set $b = \Phi^{-1}(1 - \alpha/2) = z_{1-\alpha/2}$.

CI for μ , Known Variance

$$\left(\bar{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

is a $(1 - \alpha)$ confidence interval for μ . The half-width $z_{1-\alpha/2} \sigma/\sqrt{n}$ shrinks like $1/\sqrt{n}$.

16 Variance of a Normal Population, Known Mean

For μ known,

$$\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi_n^2$$

is a pivot. Choose a, b such that

$$P(a < \chi_n^2 < b) = 1 - \alpha \quad (\text{typically with } \alpha/2 \text{ in each tail}).$$

Then

$$\left(\frac{1}{b} \sum_{i=1}^n (X_i - \mu)^2, \quad \frac{1}{a} \sum_{i=1}^n (X_i - \mu)^2 \right)$$

is a $(1 - \alpha)$ CI for σ^2 .

17 Variance of a Normal Population, Unknown Mean

The sample variance $S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$ is unbiased for σ^2 , and

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

independently of $\bar{X}$. The corresponding CI is

$$\left(\frac{(n-1)S^2}{b_{n-1}}, \frac{(n-1)S^2}{a_{n-1}} \right),$$

where a_{n-1}, b_{n-1} are the appropriate quantiles of χ_{n-1}^2 . Note that the loss of one degree of freedom widens the interval slightly compared to the known-mean case, reflecting the cost of estimating μ from the same data.

18 Mean of a Normal Population, Unknown Variance

Story: “Student” at Guinness

“That ancient and honorable firm, the Guinness Brewing Company of Dublin, Ireland, began the twentieth century with an investment in science.” Young Lord Guinness, the new heir, hired leading graduates in chemistry from Oxford and Cambridge. In 1899 he recruited a 23-year-old Oxford graduate with a combined degree in chemistry and mathematics: William Sealy Gosset.

Gosset’s first published paper, in 1904, tackled a fermentation problem: estimating the count of live yeast cells in a jar of fluid before it was added to the mash. Too few yeast led to incomplete fermentation; too many led to bitter beer. Gosset showed under the microscope that yeast counts followed what is now known as the Poisson distribution.

In 1906 he took a year’s leave to study with Karl Pearson at the Galton Biometrical Laboratory. Pearson wanted to print Gosset’s work in *Biometrika*, but Guinness, after losing a brewing secret to a previous employee’s article, had forbidden its staff from publishing under their own names. So Gosset’s first *Biometrika* paper, like the more than thirty that followed over the next three decades, appeared under the pseudonym “Student.”

Gosset’s problem, sitting at his kitchen table at night, was that Pearson’s methods assumed large samples; experiments at Guinness might yield only ten or twenty observations. He took ratios of estimated means to estimated standard deviations, tabulated their distribution by hand, and matched the result against Pearson’s family of skew distributions. The 1908 *Biometrika* paper “The Probable Error of the Mean” is the origin of what we now call Student’s *t*-distribution. (Gosset used the letter *z*; later textbook writers introduced the letter *t*.) It was Fisher, then a young Cambridge wrangler, who proved Gosset’s results rigorously using multidimensional geometry. The Guinness family did not learn of “Student”’s identity until after Gosset’s heart attack in 1937.

Source: Salsburg (2001), *The Lady Tasting Tea*, ch. 3, pp. 25–32.

Combining the two pivots,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \text{and} \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \quad (\text{independently}),$$

the ratio

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{\frac{(n-1)S^2/\sigma^2}{n-1}}} \sim T_{n-1}$$

is a pivot. The unknown σ cancels: it is a *nuisance parameter*.

CI for μ , Unknown Variance (t Interval)

$$\left(\bar{X} - t_{n-1, 1-\alpha/2} \frac{S}{\sqrt{n}}, \quad \bar{X} + t_{n-1, 1-\alpha/2} \frac{S}{\sqrt{n}} \right)$$

is a $(1 - \alpha)$ CI for μ . The t quantile is larger than the corresponding z quantile at small n , widening the interval to honestly reflect the uncertainty added by estimating σ from the same data.

R companion: `08-cap-torque-eda.R`⁹

```
data <- read_excel("Cap Torque Data.xlsx")
data$Machine <- as.factor(data$Machine)
ggplot(data, aes(x = Torque, color = Machine)) +
  geom_histogram(binwidth = 1, fill = "gray", alpha = 0.5,
                 position = "identity")
stat.desc(data$Torque, basic = FALSE)           # overall descriptive stats
```

18.1 Selected t -Distribution Quantiles ($t_{0.975}$)

df = $n - 1$	$t_{0.975}$
1	12.71
2	4.30
3	3.18
4	2.78
5	2.57
10	2.23
20	2.086
30	2.04
60	2.00
120	1.98
∞	1.96

The t distribution converges to $N(0, 1)$ as $\text{df} \rightarrow \infty$, but at low df has substantially heavier tails to account for the additional uncertainty from estimating σ .

Practice 4: Confidence Intervals for a Manufacturing Process

⁹<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/08-cap-torque-eda.R>

A quality engineer is responsible for the torque required to remove the cap from a shampoo bottle. The target torque is 18 inch-pounds. Caps fastened too loosely fall off in shipping, while caps fastened too tightly are difficult for customers to remove. A random sample of n caps is collected from each of two injection-molding machines and their removal torques are measured.

- For each machine separately, construct a 95% and a 99% confidence interval for the mean removal torque. Interpret the intervals in plain language for a manager.
- Assuming the torque is normally distributed, compute the probability that a randomly produced cap from each machine has a torque within the tolerance range [17, 19] inch-pounds.

19 Difference in Means of Two Normal Populations

19.1 Known Variances

For independent samples $X_1, \dots, X_{n_1} \sim N(\mu_1, \sigma_1^2)$ and $Y_1, \dots, Y_{n_2} \sim N(\mu_2, \sigma_2^2)$,

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

A $(1 - \alpha)$ CI for $\mu_1 - \mu_2$ is

$$\left((\bar{X} - \bar{Y}) \pm z_{1-\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right).$$

19.2 Common Unknown Variance

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$ but unknown, the pooled sample variance

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

satisfies $(n_1 + n_2 - 2) S_p^2 / \sigma^2 \sim \chi_{n_1+n_2-2}^2$, independently of $\bar{X} - \bar{Y}$. The pooled-variance two-sample t statistic

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim T_{n_1+n_2-2}.$$

Pooled-Variance Two-Sample t CI

Under the common-variance assumption $\sigma_1^2 = \sigma_2^2$,

$$\left((\bar{X} - \bar{Y}) \pm t_{n_1+n_2-2, 1-\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}\right)$$

is a $(1 - \alpha)$ CI for $\mu_1 - \mu_2$. Combining the two samples into one pooled variance estimate buys $n_1 + n_2 - 2$ degrees of freedom, more than either sample alone.

19.3 Unequal Unknown Variances

When $\sigma_1^2 \neq \sigma_2^2$ and both are unknown, the variances do not cancel from numerator and denominator. The Welch–Satterthwaite procedure provides an approximate t statistic with adjusted degrees of freedom (out of scope here).

20 Difference in Two Population Proportions

For $Y_1 \sim \text{Binomial}(n_1, p_1)$ and $Y_2 \sim \text{Binomial}(n_2, p_2)$ independent, the sample proportions $\hat{p}_k = Y_k/n_k$ satisfy

$$E(\hat{p}_k) = p_k, \quad \text{Var}(\hat{p}_k) = \frac{p_k(1-p_k)}{n_k}.$$

For large n_1, n_2 ,

$$\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \stackrel{d}{\approx} N(0, 1),$$

giving a normal-approximation CI for $p_1 - p_2$. To remove the dependence on the unknown p_k in the standard error, one of three substitutions is typically used:

- **Plug-in (Wald):** replace p_k by $\hat{p}_k$.
- **Conservative:** replace $p_k(1-p_k)$ by its maximum $1/4$.
- **Pooled estimator (used under $H_0 : p_1 = p_2$):**

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}, \quad \widehat{\text{SE}} = \sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}.$$

The plug-in standard error gives a tighter (less wasteful) interval than the conservative bound when the true proportions are far from $1/2$, e.g. for low click-through or rare-disease rates.

21 Ratio of Variances of Two Normal Populations

If $X \sim \chi_n^2$ and $Y \sim \chi_m^2$ are independent,

$$\frac{X/n}{Y/m} \sim F(n, m).$$

Hence

$$\frac{(n_1 - 1) S_1^2 / \sigma_1^2}{(n_2 - 1) S_2^2 / \sigma_2^2} \cdot \frac{n_2 - 1}{n_1 - 1} = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F(n_1 - 1, n_2 - 1),$$

which can be inverted to a $(1 - \alpha)$ CI for σ_2^2 / σ_1^2 using F quantiles.

22 Relationship Between T , χ^2 , and F Distributions

Relationships Among N , χ^2 , T , and F

If $Z \sim N(0, 1)$ and $Y \sim \chi_n^2$ independently, then

$$\frac{Z}{\sqrt{Y/n}} \sim T(n), \quad \left(\frac{Z}{\sqrt{Y/n}} \right)^2 \sim F(1, n).$$

More generally, the ratio of two independent rescaled χ^2 variables is F -distributed:

$$\frac{\chi_n^2/n}{\chi_m^2/m} \sim F(n, m).$$

This last identity is the foundation of ANOVA's F test.

A confidence interval is a passive description of where the parameter probably lies. Part V switches to the active question: given a candidate value of the parameter, do the data provide enough evidence to reject it? The framework distinguishes the two kinds of mistakes (rejecting a true null vs. accepting a false one), formalizes power, and uses the same pivots we built for CIs to construct test statistics.

Part V

Hypothesis Testing

Weeks 4–6 · What We Cover in This Part

This part is the decision-theoretic counterpart of the previous one: instead of describing where the parameter lies, we test whether the data are inconsistent with a null value. We introduce the setup and language of H_0 vs H_1 , define Type I and Type II errors, and study the power function. We then design n and c to hit chosen error rates and work through significance levels and p -values. We finish with one- and two-sample tests, the two-proportion z -test, and a set of worked examples (vaccine efficacy, Fisher’s exact test) that recur in the rest of the course.

Textbook: Montgomery, Ch. 4, pp. 111–143.

23 Setup

A hypothesis test is a procedure for deciding, on the basis of data, whether to reject a default “null” description of the world.

- **Null hypothesis H_0 :** the default setting (no effect, no difference).
- **Alternative H_1 :** a competing claim of interest.
- **Test statistic:** a function of the data used to decide.
- **Critical region:** the set of data values for which we reject H_0 .

A canonical one-sided test for the mean of a normal population $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ is

$$H_0 : \mu \leq \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0, \quad \text{reject } H_0 \text{ if } \bar{X} > c.$$

24 Type I and Type II Errors

	H_0 true	H_0 false
Reject H_0	Type I error (false positive)	Correct (true positive)
Fail to reject	Correct (true negative)	Type II error (false negative)

Sensitivity and specificity. For a binary classifier (H_0 = “no condition”), the sensitivity (true positive rate) and specificity (true negative rate) are

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}.$$

Sensitivity is one minus the Type II error rate, and specificity is one minus the Type I error rate.

Practice 5: Binary Signal Transmission

A digital channel transmits the bits 0 and 1 as voltages +1 and +7 respectively. During transmission a noise voltage $N \sim N(0, 1)$ is added to the signal. The receiver interprets a received voltage above 5 as a 1, below 3 as a 0, and drops the message if the received voltage is between 3 and 5.

- (a) A Type I error is calling a transmitted 0 a 1. Compute its probability.
- (b) A Type II error is calling a transmitted 1 a 0. Compute its probability.
- (c) Compute the probability that the message is dropped.

Justify all your answers mathematically.

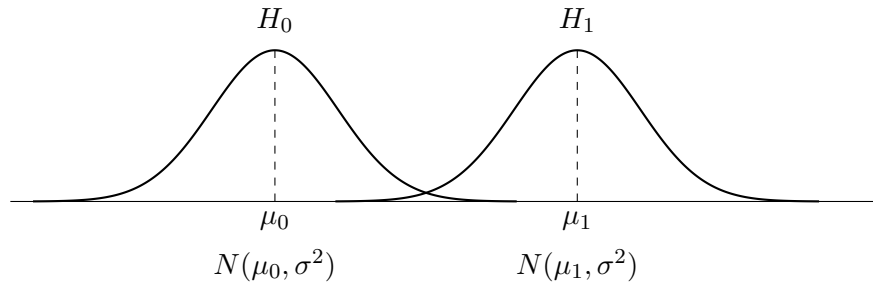
25 Power Function

The *power function* is the probability of rejecting H_0 as a function of the parameter:

$$K(\mu) = P_\mu(\text{reject } H_0).$$

For the canonical one-sided test,

$$K(\mu) = P_\mu(\bar{X} > c) = 1 - \Phi\left(\frac{c - \mu}{\sigma/\sqrt{n}}\right).$$



The smaller the overlap of the two distributions (e.g. a larger effect size $\mu_1 - \mu_0$ or a smaller $\sigma/\sqrt{n}$), the higher the power.

26 Designing n and c

To control both error rates, we set:

- $\alpha = K(\mu_0)$, the Type I error rate at the boundary,
- $\beta = 1 - K(\mu_1)$, the Type II error rate at a specified alternative $\mu_1 > \mu_0$.

Solving

$$1 - \Phi\left(\frac{c - \mu_0}{\sigma/\sqrt{n}}\right) = \alpha \quad \text{and} \quad 1 - \Phi\left(\frac{c - \mu_1}{\sigma/\sqrt{n}}\right) = 1 - \beta$$

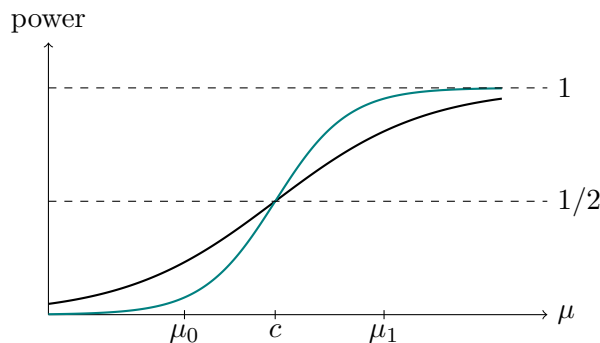
yields the critical value and the required sample size simultaneously.

Sample Size and Critical Value for a One-Sided z Test

$$c = \mu_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}}, \quad n \geq \left(\frac{(z_{1-\alpha} + z_{1-\beta}) \sigma}{\mu_1 - \mu_0} \right)^2.$$

The sample size grows like $1/(\mu_1 - \mu_0)^2$: distinguishing closer alternatives requires quadratically more data.

Decreasing c increases α but decreases β . The two error rates are not free parameters of nature: they trade off, and the lever for relaxing the trade-off is sample size.



The teal curve has larger sample size: it is steeper, achieving higher power at the alternative while maintaining the same false positive rate at μ_0 .

27 Significance Level and p -Values

The *significance level* of a test is the maximum Type I error probability, $\sup_{\mu \in H_0} K(\mu)$. The *p-value* is a data-dependent quantity: the smallest significance level at which the observed data would lead to rejection. Equivalently, for a sharp null $\mu = \mu_0$, the *p-value* is the probability of observing a test statistic at least as extreme as the one observed, under H_0 .

Sharp nulls and CIs. To test a sharp null $\mu = \mu_0$, one can construct a $(1 - \alpha)$ confidence interval for μ and reject if it does not contain μ_0 . This is the duality between confidence intervals and tests.

Story: The Bayesian Heresy

The p -value answers a question of the form, “how often, in repeated experiments under the null, would I see data this extreme?” It does *not* answer, “what is the probability that the null is true given the data I saw.” The second question requires Bayes’s theorem and a prior probability for the null, and the two questions can give very different answers.

In the late eighteenth century, the Reverend Thomas Bayes noticed an inner symmetry in the formulas for conditional probability: from $P(\text{data} \mid \text{hypothesis})$ one can recover $P(\text{hypothesis} \mid \text{data})$ if one supplies a prior. “Bayes put these calculations aside. They were found among his papers when he died, and were published posthumously.”

For most of the twentieth century, this “inverse probability” was treated with suspicion. Fisher was once accused of using it at the Royal Statistical Society and “stoutly defended himself against such a terrible charge.” Neyman’s original confidence-interval paper relied on Bayesian machinery, and his second paper showed how to reach the same answer without it.

The objection was philosophical: the Pearsonian revolution rested on parameters being fixed though unknown, and treating them as random felt incoherent.

By the end of the twentieth century the objection had softened: “over half the articles that appear in journals like the *Annals of Statistics* and *Biometrika* now make use of Bayesian methods.” The Bayesian scheme,

prior probability $\longrightarrow$ data $\longrightarrow$ posterior probability,

updates an existing belief in light of new data. The pragmatic point for hypothesis testing is that a small frequentist p -value can still leave the posterior probability of the null large, if one starts with a strong prior in favor of the null. The two frameworks ask different questions, and they need not agree.

Source: Salsburg (2001), The Lady Tasting Tea, ch. 13, pp. 125–129.

Practice 6: One Toss to Detect a Biased Coin

A coin is suspected of being twice as likely to land heads as tails ($p = 2/3$ rather than $p = 1/2$). The coin is to be tossed exactly once. Provide a (powerful!) test of $H_0 : p = 1/2$ at significance level $\alpha = 0.10$, and compute its power $1 - \beta$. Justify your answer mathematically. Discuss why a single toss imposes a hard ceiling on the achievable power and what additional tools (such as randomized tests) would be required to push past it.

Practice 7: A 400-Toss Test and Its p -Value

A coin is tossed $n = 400$ times. Let X be the number of heads, so $X \sim \text{Binomial}(400, p)$. We test $H_0 : p = 1/2$ against $H_1 : p > 1/2$ using the standardized statistic $Y = (X - 200)/10$, which is approximately $N(0, 1)$ under H_0 . We reject H_0 if $Y > c$.

- (a) Find c such that the significance level is $\alpha = 0.10$.
- (b) State the rejection region in terms of the original count X .
- (c) Suppose we observe $X = 220$. Find the smallest α at which we would reject H_0 given this observation. This is the (in)famous p -value.

28 One-Sample Tests

One-Sample z and t Tests

For testing $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$:

$$(\text{known } \sigma) \quad Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \stackrel{H_0}{\sim} N(0, 1), \quad \text{reject if } |Z| > z_{1-\alpha/2}.$$

$$(\text{unknown } \sigma) \quad T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \stackrel{H_0}{\sim} T_{n-1}, \quad \text{reject if } |T| > t_{n-1, 1-\alpha/2}.$$

29 Two-Sample Tests

Story: The Intent to Treat

In the early 1980s, Richard Peto, one of the leading biostatisticians in Great Britain, was analyzing the results of several cancer clinical trials. Following Fisher's dictates of experimental design, each trial had randomized a group of patients between different experimental treatments. The two-sample comparison should have been straightforward.

It was not. Many of the patients had not received the treatments to which they were randomized. "These were human beings, suffering from painful and, in many cases, terminal disease." Treating physicians, faced with a patient failing on the assigned arm, felt impelled to switch. "Blind following of an arbitrary treatment, without considering the patient's needs and responses, would have been unethical."

The usual practice up to that point was to drop, from the analysis, any patient who had not stayed on the randomized treatment. Peto pointed out that this could lead to serious bias. Imagine a placebo versus an active treatment: the patients who fail on placebo will be switched to the active treatment and dropped, leaving only the placebo responders in the placebo column. "The placebo will be found to be as effective (or perhaps even more so) than the active treatment."

Peto's solution was to analyze every patient under the arm they had been *randomized* to, regardless of what they actually received. "If a patient randomized to treatment A never took a single pill from treatment A but was hospitalized and put on alternative therapies immediately after entering the study, that patient was analyzed as a treatment A patient." This came to be called the *intent-to-treat* method, and is now enshrined in medical doctrine as the analysis of record for clinical trials that test public-health policy. Peto himself was careful to add a caveat: the method "can only be used if the analysis finds that they differ in effect." Failing to find a difference does not prove that the treatments are equivalent.

Source: Salsburg (2001), *The Lady Tasting Tea*, ch. 27, pp. 275–278.

Two-Sample z and t Tests

For testing $H_0 : \mu_1 = \mu_2$:

$$(\text{known } \sigma_1, \sigma_2) \quad Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \stackrel{H_0}{\sim} N(0, 1).$$

$$(\text{common unknown } \sigma, \text{ pooled variance } S_p^2) \quad T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{1/n_1 + 1/n_2}} \stackrel{H_0}{\sim} T_{n_1+n_2-2}.$$

Reject at level α if the absolute test statistic exceeds the corresponding two-sided critical value.

Practice 8: Comparing Two Engines

The viscosity of motor oil is known to drift after extended use because of high engine temperatures. An experiment compares the resulting viscosity from two engines. Engine 1 contributes n_1 samples with mean $\bar{X}$ and sample standard deviation s_X . Engine 2 contributes n_2 samples with mean $\bar{Y}$ and sample standard deviation s_Y . Test the hypothesis that the two engines produce the same population mean viscosity at significance level $\alpha = 0.10$.

- Compute the test statistic and decide the test using the general two-sample procedure (unequal variances), reporting the appropriate t statistic and degrees of freedom.
- Repeat using the pooled-variance procedure under the assumption that the two engines share a common variance.

Justify all your answers mathematically.

*R companion: 09-t-tests-and-ci.R*¹⁰

```
t.test(M1$Torque, conf.level = 0.95)           # one-sample CI for the mean
t.test(M1$Torque, M2$Torque, conf.level = 0.95) # two-sample comparison
t.test(votes$diff_log_votes, conf.level = 0.99) # paired-style on log-difference
```

29.1 Two-Proportion z Test

For $H_0 : p_1 = p_2$ vs. $H_1 : p_1 \neq p_2$, the natural test statistic uses the pooled estimator $\hat{p} = (Y_1 + Y_2)/(n_1 + n_2)$.

¹⁰<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/09-t-tests-and-ci.R>

Two-Proportion z Test

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \stackrel{H_0}{\sim} N(0,1) \quad (\text{asymptotically}), \quad \text{reject if } |Z| > z_{1-\alpha/2}.$$

The pooled standard error uses $\hat{p}$ rather than the separate $\hat{p}_k$, since H_0 asserts a single common proportion.

30 Worked Examples

Example 30.1 (Vaccine efficacy for a rare disease). A vaccine is administered to n animals in a population with disease prevalence p , and after the observation period none develop the disease. Under the null that the vaccine has no effect,

$$P(S_n = 0) = (1 - p)^n.$$

For $p = 0.01, n = 10$: $P(S_n = 0) \approx 0.90$, no evidence of efficacy. For $p = 0.5, n = 10$: $P(S_n = 0) \approx 0.001$, strong evidence of efficacy.

Lesson: establishing efficacy against rare outcomes requires correspondingly larger samples.

Example 30.2 (Coin flip with limited evidence). A coin is flipped n times under $H_0 : p = 1/2$. A reasonable two-sided test rejects when the outcome is all heads or all tails. The smallest n for which this test has level at most 0.05 is the smallest n such that $(1/2)^{n-1} < 0.05$, i.e. $n^* = \lceil \log_2 40 \rceil = 6$.

Lesson: with very few flips, the discreteness of the outcome space prevents any deterministic test of nontrivial level. Randomized tests are required to attain arbitrary α .

Example 30.3 (A/B testing of a website button). A randomized experiment compares an old and a new button design. Of n_1 visitors shown the old design, Y_1 click. Of n_2 visitors shown the new design, Y_2 click. To test $H_0 : p_1 = p_2$, apply the two-proportion z test and reject at level 0.05 if $|Z| > 1.96$. The plug-in standard error generally yields a tighter test than the conservative $1/4$ bound when the click-through rates are far from $1/2$.

Example 30.4 (Cumulative “at least as extreme” probability). For a Fisher-style exact test based on the hypergeometric, observing i successes is evaluated against

$$P(X \geq i) = \sum_{j \geq i} \frac{\binom{K}{j} \binom{N-K}{n-j}}{\binom{N}{n}},$$

the probability of observing i or any more extreme outcome. The p -value is always a tail (cumulative) probability, never the point probability of the observed value alone.

*R companion: `10-grouped-summaries-facets.R`*¹¹

```
votes %>%
  group_by(state_name) %>%
  summarise(count = n(), mean_dem = mean(votes_dem), sd_dem = sd(votes_dem))
```

¹¹<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/10-grouped-summaries-facets.R>

```
ggplot(votes, aes(votes_dem)) +  
  geom_histogram(aes(y = stat(density)), bins = 15) +  
  geom_density() + facet_grid(. ~ state_name)
```

Two-sample tests compare two groups. But experiments routinely involve more than two groups (three operators, four treatments) or more than one factor (temperature *and* pressure *and* time). Part VI generalizes the comparison through the analysis of variance, partitioning the total variation into a between-group component (which the factor explains) and a within-group component (which it does not). The F statistic asks whether the explained portion is too large to be attributed to chance.

Part VI

Analysis of Variance

Week 7 · What We Cover in This Part

This part generalizes two-sample testing to many groups, with the variability decomposed into a between-group component (which a factor explains) and a within-group component (which it does not). We work through one-way ANOVA: the setup, within and between variability, the sum-of-squares decomposition, the distributions under the null, the F statistic, and the ANOVA table. We then extend to two-way ANOVA with an interaction term, covering the model, the sums of squares, and the full two-way ANOVA table.

Textbook: Montgomery, Ch. 4, pp. 143–152, and Ch. 13, p. 519, Example 13.5.

31 One-Way ANOVA

31.1 Setup

Suppose k treatment groups produce observations

$$y_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n_i,$$

with group means $\bar{y}_{i\cdot}$ and grand mean $\bar{y}_{\cdot\cdot}$. Let $n_T = n_1 + \dots + n_k$.

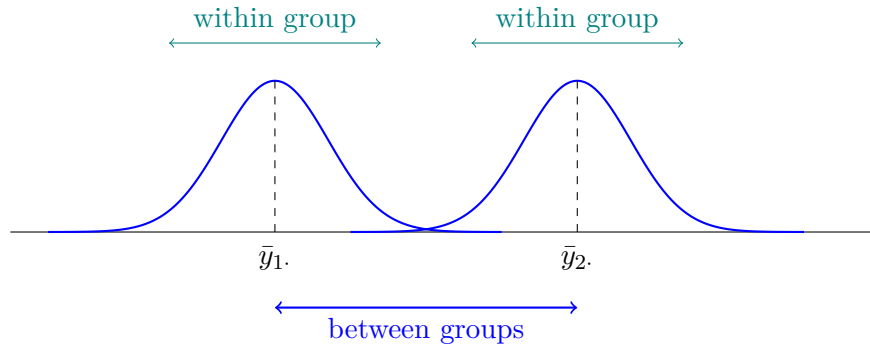
Under the fixed-effects model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

the null hypothesis of no treatment effect is

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_k.$$

31.2 Within and Between Variability



31.3 Sum of Squares Decomposition

Adding and subtracting the group mean inside each squared deviation,

$$\sum_{i,j} (y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot})^2 + \sum_i n_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + 2 \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot})(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}).$$

The cross term vanishes because $\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\cdot}) = 0$ for each i .

One-Way ANOVA Sum-of-Squares Identity

$$\underbrace{\sum_{i,j} (y_{ij} - \bar{y}_{..})^2}_{SS_T} = \underbrace{\sum_{i,j} (y_{ij} - \bar{y}_{i.})^2}_{SS_W = SS_E} + \underbrace{\sum_i n_i (\bar{y}_{i.} - \bar{y}_{..})^2}_{SS_B = SS_{\text{Treatment}}}$$

Total variation splits cleanly into a within-group component (noise) and a between-group component (signal).

31.4 Distributions Under the Null and the F-Statistic

Under H_0 ,

$$\frac{1}{\sigma^2} SS_W \sim \chi_{n_T - k}^2, \quad \frac{1}{\sigma^2} SS_B \sim \chi_{k-1}^2, \quad \frac{1}{\sigma^2} SS_T \sim \chi_{n_T - 1}^2,$$

with SS_W and SS_B independent.

One-Way ANOVA F Statistic

With $MS_B = SS_B/(k-1)$ and $MS_W = SS_W/(n_T - k)$,

$$F = \frac{MS_B}{MS_W} \stackrel{H_0}{\sim} F(k-1, n_T - k), \quad \text{reject } H_0 \text{ if } F > f_{1-\alpha}(k-1, n_T - k).$$

Under H_0 , $E(MS_B) = E(MS_W) = \sigma^2$ so $F \approx 1$. Genuine treatment effects inflate MS_B and push F to the right.

31.5 ANOVA Table

Source	df	SS	MS	F
Treatment / Between	$k - 1$	$\sum_i n_i (\bar{y}_{i.} - \bar{y}_{..})^2$	$SS_B/(k-1)$	MS_B/MS_W
Error / Within	$n_T - k$	$\sum_{i,j} (y_{ij} - \bar{y}_{i.})^2$	$SS_W/(n_T - k)$	
Total	$n_T - 1$	$SS_B + SS_W$		

Remark 31.1 (Two-sample test as an ANOVA special case). For $k = 2$, $F = T^2$ where T is the pooled-variance two-sample t statistic. This is consistent with $T_m^2 \sim F(1, m)$.

Remark 31.2 (Residual degrees of freedom). The error degrees of freedom $n_T - k$ control the variance of MS_W . With more residual df, MS_W is estimated more precisely and the test has more power, holding effect size fixed. This is one reason balanced designs with adequate replication are preferred.

Practice 9: p -Value Behavior Under SS Changes

Suppose we run a one-way ANOVA on k treatments and observe a p -value α_1 .

- We rerun the experiment and obtain a new sample with the same SS_E but a larger $SS_{\text{Treatment}}$. Will the p -value of the new test be larger or smaller than α_1 ? Justify mathematically.
- We rerun the experiment again and obtain a new sample with the same $SS_{\text{Treatment}}$

but a larger SS_E . Will the p -value of this third test be larger or smaller than α_1 ? Justify mathematically.

Reflect. If experimenters could publish only when their p -value is below 0.05, can they keep rerunning the experiment until a significant outcome appears? Connect this to the multiple-testing problem discussed below.

R companion: `11-one-way-anova.R`¹²

```
my_data <- PlantGrowth
ggplot(my_data, aes(x = group, y = weight, fill = group)) + geom_boxplot()
result_aov <- aov(weight ~ group, data = my_data)
summary(result_aov)
```

32 Two-Way ANOVA

32.1 Model

With factors A at a levels and B at b levels and n replicates per treatment combination,

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

for $i = 1, \dots, a$, $j = 1, \dots, b$, $k = 1, \dots, n$. The grand total $n_T = abn$. The interaction term $(\tau\beta)_{ij}$ captures the extent to which the effect of A depends on the level of B and vice versa.

32.2 Sums of Squares

Two-Way ANOVA Sum-of-Squares Identity

$$\begin{aligned} SS_A &= bn \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2, & SS_B &= an \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2, \\ SS_{AB} &= n \sum_{i,j} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2, & SS_E &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2, \end{aligned}$$

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E.$$

The total variation splits into two main effects, an interaction, and a residual, each estimable from a single dataset.

¹²<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/11-one-way-anova.R>

32.3 Two-Way ANOVA Table

Source	df	SS	MS	F
A	$a - 1$	SS_A	$SS_A/(a - 1)$	MS_A/MS_E
B	$b - 1$	SS_B	$SS_B/(b - 1)$	MS_B/MS_E
$A:B$	$(a - 1)(b - 1)$	SS_{AB}	$SS_{AB}/[(a - 1)(b - 1)]$	MS_{AB}/MS_E
Error	$ab(n - 1)$	SS_E	$SS_E/[ab(n - 1)]$	
Total	$abn - 1$	SS_T		

Each F ratio is compared against an F quantile with the appropriate degrees of freedom. The null of zero main or interaction effect is rejected at level α when the F ratio exceeds $f_{1-\alpha}$.

R companion: `12-two-way-anova.R`¹³

```
result_aov <- aov(adhesion_force ~ method + type + method:type, data = my_data)
summary(result_aov)
ggplot(my_data, aes(x = type, y = adhesion_force, fill = method)) +
  geom_boxplot() + stat_summary(fun = mean, geom = "line", aes(group = method))
```

Two-way ANOVA shows what is possible when two factors are varied together. Part VII pushes further: many factors at once, with each factor restricted to two levels, and with a careful accounting of which interaction effects can still be estimated when the full grid becomes too expensive to run. Blocking, confounding, and fractional factorials are the tools that make industrial experimentation feasible.

¹³<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/12-two-way-anova.R>

Part VII

Factorial and Fractional Factorial Designs

Week 9 · What We Cover in This Part

This part shows how to plan, rather than merely analyze, a multi-factor experiment. We explain why restricting each factor to two levels buys a great deal of simplification, then work through the geometry, design matrix, effect estimates, sums of squares, and ANOVA table for the 2^2 , 2^3 , and general 2^k designs. We introduce the sparsity-of-effects principle, treat blocking and confounding when not all 2^k runs can be performed under uniform conditions, and turn to fractional factorial designs (2^{3-1} , the general 2^{k-p} fraction, and 2^{5-1}). We close with center points for curvature detection and Pareto plots for effect screening.

Textbook: Montgomery, Ch. 13, pp. 523–533 and pp. 549–553, and Ch. 13–14.

33 Why Two Levels?

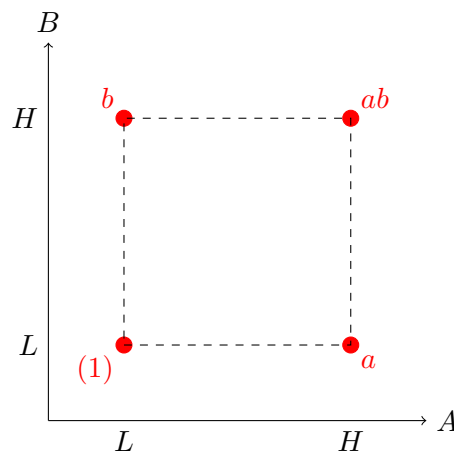
When studying many factors simultaneously, restricting each factor to two levels (coded “low” for – and “high” for +) leads to dramatic simplifications:

- Every main effect, two-way interaction, etc., has exactly 1 degree of freedom.
- Effects can be estimated by simple linear contrasts of the cell totals.
- The ANOVA table for a 2^k design has a uniform structure regardless of k .

The price is that, with only two levels per factor, one cannot detect curvature in the response surface. Center points (Section ??) are an inexpensive remedy.

34 The 2^2 Design

34.1 Geometry and Notation



In Yates' notation, the labels (1), a , b , ab stand for the *sums* of n replicate measurements obtained at each treatment combination. The lowercase letters present indicate which factors are at the high level.

34.2 Design Matrix

Run	A	B	AB
(1)	−	−	+
a	+	−	−
b	−	+	−
ab	+	+	+

The signs in the AB column are the products of the corresponding A and B signs.

34.3 Effect Estimates

The effect of A at low B is $(a - (1))/n$, and at high B it is $(ab - b)/n$. Averaging,

$$A = \frac{(a - (1)) + (ab - b)}{2n} = \frac{-(1) + a - b + ab}{2n}.$$

Similarly,

$$B = \frac{-(1) - a + b + ab}{2n}, \quad AB = \frac{(1) - a - b + ab}{2n}.$$

Equivalently, each effect equals the inner product of its column of signs with the cell totals, divided by $2n$.

34.4 Sums of Squares

For each effect $E \in \{A, B, AB\}$, with *contrast* = numerator of E above (a signed sum of cell totals),

$$SS_E = \frac{(\text{contrast})^2}{2^k n} = \frac{(\text{contrast})^2}{4n} \quad (k = 2).$$

34.5 ANOVA Table

Source	df	SS	MS	F
A	1	SS_A	SS_A	MS_A/MSE
B	1	SS_B	SS_B	MS_B/MSE
AB	1	SS_{AB}	SS_{AB}	MS_{AB}/MSE
Error	$4(n - 1)$	SS_E	$SS_E/[4(n - 1)]$	
Total	$4n - 1$	SS_T		

Each F is compared with $f_{1-\alpha}(1, 4n - 4)$. The error sum of squares can be obtained either directly from $\sum (y_{ijk} - \bar{y}_{ij.})^2$ or by subtraction: $SS_E = SS_T - SS_A - SS_B - SS_{AB}$.

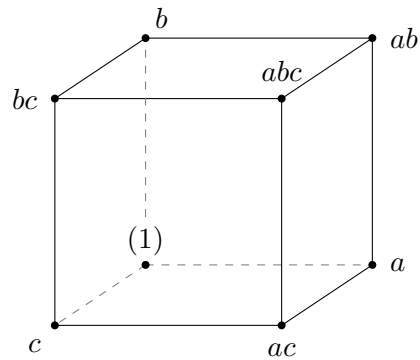
Practice 10: A 2^2 Foundry Experiment

A foundry has been asked to lower the carbon content (factor A) and increase the chromium content (factor B) of a product. The effect on hardness and strength is unknown, so you design a 2^2 factorial with $n = 3$ replicates. Carbon takes levels 0.10% (low) and 0.04% (high). Chromium takes levels 24% (low) and 28% (high). The response variables are Rockwell hardness and tensile strength.

- (a) Identify the factors and the response variables of the experiment.
- (b) Given replicate measurements for hardness and strength at each of the four corners, fit the 2^2 model and compute the main and interaction effects.
- (c) Report which effects are statistically significant at the 5% level using the ANOVA table for a 2^2 design with three replicates.
- (d) Critique the experimental design that was prescribed: what improvements would you suggest?

35 The 2^3 Design

35.1 Geometry



35.2 Design Matrix

Run	<i>A</i>	<i>B</i>	<i>C</i>	<i>AB</i>	<i>AC</i>	<i>BC</i>	<i>ABC</i>
(1)	−	−	−	+	+	+	−
<i>a</i>	+	−	−	−	−	+	+
<i>b</i>	−	+	−	−	+	−	+
<i>ab</i>	+	+	−	+	−	−	−
<i>c</i>	−	−	+	+	−	−	+
<i>ac</i>	+	−	+	−	+	−	−
<i>bc</i>	−	+	+	−	−	+	−
<i>abc</i>	+	+	+	+	+	+	+

35.3 Effects

With $k = 3$, each effect E equals (column-of-signs $\cdot$ cell totals)/($2^{k-1}n$) = $(\cdot)/(4n)$. Explicitly:

$$\begin{aligned} A &= \frac{1}{4n} [-(1) + a - b + ab - c + ac - bc + abc], \\ B &= \frac{1}{4n} [-(1) - a + b + ab - c - ac + bc + abc], \\ C &= \frac{1}{4n} [-(1) - a - b - ab + c + ac + bc + abc], \\ AB &= \frac{1}{4n} [(1) - a - b + ab + c - ac - bc + abc], \\ AC &= \frac{1}{4n} [(1) - a + b - ab - c + ac - bc + abc], \\ BC &= \frac{1}{4n} [(1) + a - b - ab - c - ac + bc + abc], \\ ABC &= \frac{1}{4n} [-(1) + a + b - ab + c - ac - bc + abc]. \end{aligned}$$

For each, $SS_E = (\text{contrast})^2 / (2^k n) = (\cdot)^2 / (8n)$.

35.4 ANOVA Table

Source	df	SS	MS	F
A	1	SS_A	MS_A	MS_A/MSE
B	1	SS_B	MS_B	MS_B/MSE
C	1	SS_C	MS_C	MS_C/MSE
AB	1	SS_{AB}	MS_{AB}	MS_{AB}/MSE
AC	1	SS_{AC}	MS_{AC}	MS_{AC}/MSE
BC	1	SS_{BC}	MS_{BC}	MS_{BC}/MSE
ABC	1	SS_{ABC}	MS_{ABC}	MS_{ABC}/MSE
Error	$8(n-1)$	SS_E	$SS_E/[8(n-1)]$	
Total	$8n-1$	SS_T		

R companion: [13-2k-factorial-basic.R](#)¹⁴

```
base <- expand.grid(A = c(-1, 1), B = c(-1, 1), C = c(-1, 1))
data <- rbind(base, base); data$r <- rep(1:4, 4) # replicated 2^3
model <- lm(r ~ A * B * C, data = data)
summary(aov(model))
predict(model, data.frame(A = 2, B = 1, C = 0)) # extrapolate
```

36 General 2^k Design

A 2^k design has

$$2^k - 1 = k + \binom{k}{2} + \binom{k}{3} + \cdots + \binom{k}{k}$$

estimable effects, each with 1 degree of freedom.

¹⁴<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/13-2k-factorial-basic.R>

Effect and SS Formulas for the 2^k Design

For each effect E in a 2^k design with n replicates per cell,

$$(\text{effect}) = \frac{\text{contrast}}{2^{k-1}n}, \quad SS_E = \frac{(\text{contrast})^2}{2^k n},$$

where the contrast is the inner product of E 's sign column with the cell totals. Every effect is estimated by a single linear contrast, regardless of k .

36.1 Sparsity-of-Effects Principle

In practice, systems are typically dominated by main effects and low-order interactions, while high-order interactions tend to be small. This empirical regularity is the basis for fractional factorial designs, which save runs by sacrificing the ability to estimate certain (presumed-small) higher-order interactions cleanly.

Practice 11: A 2^4 Soft-Drink Experiment

A beverage company is reformulating a soft drink. Four factors are thought to affect taste: sweetener type (A), syrup-to-water ratio (B), carbonation level (C), and serving temperature (D). Each factor has two levels. At each of the $2^4 = 16$ corners of the design, twenty tasters score the beverage from 1 to 10, and the total score is recorded. The experiment is replicated twice, for a total of 32 observations.

- (a) Identify the factors and the response variable.
- (b) Set up the 2^4 design matrix in standard order and assign the experimental data to it.
- (c) Estimate the four main effects and the six two-way interactions, and rank them by absolute magnitude on a Pareto chart.
- (d) State which effects are statistically significant and recommend a beverage formulation.

37 Blocking and Confounding

Often it is impossible to perform all 2^k runs under perfectly homogeneous conditions: the experiment may span multiple days, shifts, batches of raw material, or operators. *Blocking* groups runs into homogeneous subsets to absorb the unwanted nuisance variation. The cost is that one effect is sacrificed by being *confounded* (aliased) with the block effect.

37.1 Blocking in a 2^2 Design

Suppose the four runs cannot be carried out in a single day. Splitting the runs $\{(1), ab\}$ on day 1 and $\{a, b\}$ on day 2 confounds AB with the day effect:

$$AB = \frac{1}{2n} [(1) + ab - a - b].$$

Whatever this quantity estimates includes both the genuine AB interaction and the block (day) effect, and the two cannot be separated. However, the same arrangement leaves A and B *unconfounded*:

$$A = \frac{1}{2n} [- (1) + a - b + ab], \quad B = \frac{1}{2n} [- (1) - a + b + ab].$$

In each, day 1 runs and day 2 runs appear with both signs, so the day effect cancels. The principle generalizes: *the effect whose sign pattern matches the block pattern is the one that gets confounded*.

37.2 Blocking in a 2^3 Design

To run a 2^3 in two blocks of four, sacrifice the highest-order interaction ABC :

$$\begin{array}{l|l} \text{Block 1 (e.g. day 1)} & (1), ab, ac, bc \\ \text{Block 2 (e.g. day 2)} & a, b, c, abc \end{array}$$

This corresponds to grouping by the sign of the ABC column. As before, A, B, C, AB, AC, BC are all estimated cleanly: the block effect is absorbed into the ABC contrast, which we are willing to sacrifice under the sparsity-of-effects assumption.

38 Fractional Factorial Designs

When k is moderately large, 2^k runs are too expensive. A $1/2^p$ *fraction* of a 2^k design, denoted 2^{k-p} , runs only 2^{k-p} treatment combinations, sacrificing the ability to estimate certain higher-order interactions cleanly.

38.1 The 2^{3-1} Design

A half-fraction of 2^3 has $2^{3-1} = 4$ runs. Choose the runs where $ABC = +1$:

$$\text{principal fraction: } \{a, b, c, abc\}, \quad \text{alternate fraction: } \{(1), ab, ac, bc\}.$$

The *defining contrast* (or generator) is $I = ABC$ in the principal fraction. Multiplying through by any effect gives its alias:

$$A \cdot I = A \cdot ABC = A^2BC = BC, \quad \text{so} \quad A \equiv BC.$$

Similarly $B \equiv AC$ and $C \equiv AB$. In the principal fraction the formulas simplify:

$$A = \frac{1}{2n} [a - b - c + abc], \quad BC = \frac{1}{2n} [a - b - c + abc],$$

which are literally identical. We cannot tell whether an apparent A effect is really A or really BC (or some combination) without an additional assumption, typically the sparsity-of-effects assumption that BC is negligible.

38.2 Generators and the General 2^{k-p} Fraction

Defining Relation and Alias Structure

A 2^{k-p} fractional factorial is determined by p *generators*, each set equal to ± 1 . The 2^p products of generators (including the identity I) form the *defining relation*, e.g. $I = ABCDE$ for 2^{5-1} . Each effect E has 2^p aliases obtained by multiplying E by every word in the defining

relation:

$$E \equiv E \cdot W \quad \text{for every word } W \text{ in the defining relation.}$$

The *resolution* of the design is the length of the shortest word in the defining relation. Higher resolution means low-order effects are aliased only with high-order ones.

A good fractional design uses high-order generators, so that low-order effects are aliased only with high-order effects: the longer the shortest word, the higher the resolution and the cleaner the estimates.

38.3 The 2^{5-1} Design

For five factors A, B, C, D, E at two levels, $2^5 = 32$ treatment combinations. A half-fraction (say $I = ABCDE$) requires only 16 runs. With $n = 3$ replicates this is 48 observations. The 5 main effects are aliased with 4-way interactions, and the $\binom{5}{2} = 10$ two-way interactions are aliased with 3-way interactions, both of which are typically negligible by the sparsity-of-effects principle.

Practice 12: Full vs. Fractional Designs with Blocking

An engineer is studying contamination in a manufacturing process and considers three factors: water volume, carbon content, and cutting speed, each at two levels.

- Construct a design matrix for a full 2^3 factorial design without replication.
- Construct a design matrix for a $1/2$ fractional factorial design without replication. Identify the defining contrast and the alias structure.
- Construct the $1/2$ fractional design with three replicates per run.
- The manager later decides to drop water volume from the study and run the resulting full 2^2 factorial with three replicates. Each run takes four hours, so the experiment will span multiple days. Discuss how to use *blocking* to absorb the day-to-day nuisance variation without confounding the main effects of carbon content and cutting speed.

39 Center Points and Curvature Detection

A pure 2^k design fits a first-order (linear) response surface in the coded factors plus interactions, and it cannot detect curvature. Adding n_c replicate runs at the geometric center of the design (each factor at its midpoint) gives a cheap test for pure quadratic curvature: under linearity of the underlying response, the average of the corner runs and the average of the center runs should be approximately equal. A significant difference between them indicates curvature in the response surface and motivates extending the design (e.g. to a central composite design with axial points to estimate quadratic terms).

40 Pareto Plots for Effect Screening

A *Pareto plot* of effects ranks the absolute values of the estimated factor effects from largest to smallest as horizontal bars. A vertical reference line at the appropriate critical value separates the few large effects (often left of the line at the top) from the many small effects (right of the line). The plot operationalizes the sparsity-of-effects principle visually and helps the experimenter focus on the most consequential factors and interactions.

R companion: [14-fractional-factorial-pareto.R](#)¹⁵

```
model <- lm(obs ~ (A + B + C + D + E)^2, data = my_data) # 2-way interactions only
summary(aov(model)); paretoPlot(model)                  # rank effects
# Build the 25-1 fraction defined by the generator I = ABCDE:
sign_table %>% filter(ABCDE == 1) %>% select(A, B, C, D, E)
```

Up to this point we have been designing experiments: creating data deliberately under controlled conditions to learn about a process. Once a process is set up and running, the focus shifts. Part VIII addresses the receiving end: incoming lots of components arrive from a supplier, and we have to decide whether to accept them on the basis of a sample. Part IX addresses the operating end: a process is running over time, and we have to decide whether it is still in control. Both parts use the probability and sampling tools we have built, but they ask different questions.

¹⁵<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/14-fractional-factorial-pareto.R>

Part VIII

Lot-by-Lot Acceptance Sampling

Weeks 10–11 · What We Cover in This Part

This part covers the receiving end of quality assurance: an incoming lot from a supplier and a sample-based decision on whether to accept it. We define what acceptance sampling is and place it between 100% inspection and no inspection at all. We then develop single-sampling plans for attributes and the operating-characteristic curve, introduce AQL, LTPD, and the producer's and consumer's risks, and extend to double-sampling plans. We survey the MIL-STD-105E (attributes) and MIL-STD-414 (variables) standards, and close with lot formation and random sampling.

Textbook: Montgomery, Ch. 15, §§1.1–1.3 and §§5.1–5.3, and Ch. 16, §§5.3, 5.6–5.7.

Story: From Bell Labs to the U.S. Military

Statistical acceptance sampling has its origin at Bell Telephone Laboratories. “Toward the end of the 1920s, Harold F. Dodge and Harry G. Romig, both of Bell Telephone Laboratories, developed statistically based acceptance sampling as an alternative to 100% inspection.” Bell Labs in the same decade was the site of Walter A. Shewhart’s 1924 control-chart memorandum and of the foundational work that grew into modern statistical quality control. “By the middle of the 1930s, statistical quality-control methods were in wide use at Western Electric, the manufacturing arm of the Bell System. However, the value of statistical quality control was not widely recognized by industry.”

World War II changed that. “Wartime experience made it apparent that statistical techniques were necessary to control and improve product quality.” Between 1940 and 1943, Bell Labs developed the forerunners of the U.S. Army’s military-standard sampling plans, the documents that grew into MIL-STD-105 (attributes) and MIL-STD-414 (variables). Training courses on statistical quality control were given to industry through the war years, and the American Society for Quality Control was founded in 1946.

“In the 1930s and 1940s, acceptance sampling was one of the major components of the field of statistical quality control, and was used primarily for incoming or receiving inspection. In more recent years, it has become typical to work with suppliers to improve their process performance through the use of SPC and designed experiments, and not to rely as much on acceptance sampling as a primary quality assurance tool.” The Dodge–Romig and MIL-STD plans that this part develops are the direct descendants of that Bell Labs work.

Source: Montgomery (2020), Introduction to Statistical Quality Control, 8th ed., §1.2 (pp. 9–12) and §15.1 (p. 591).

A lot of N items arrives from a supplier. Inspecting all N is too slow, too expensive, or destroys the items. Inspecting none risks accepting bad lots. How do we sample n items to decide whether to accept or reject the whole, and how do we quantify the risk we take on each side of that decision? The framework is formalized in the military-standard plans MIL-STD-105E (attributes, civilian counterpart ANSI/ASQC Z1.4) and MIL-STD-414 (variables, civilian counterpart ANSI/ASQC Z1.9), and in the Dodge–Romig system; this part develops the underlying machinery and surveys both.

Acceptance sampling is the middle of the road between 100% inspection (prohibitive) and no inspection (dangerous). The operating-characteristic curve $P_a(p)$ traces the probability of accepting a lot as a function of its true defective fraction p , and the steepness of that curve determines the trade-off between the *producer's risk* (rejecting a good lot) and the *consumer's risk* (accepting a bad lot). The framework connects directly to the hypergeometric and binomial distributions of Part II, the hypothesis-testing language of Part V, and the design considerations of Part VII.

41 What Acceptance Sampling Is

Purpose: lot disposition, not quality estimation. An acceptance-sampling plan does not estimate the quality of a lot. It accepts or rejects the lot. Three points are worth keeping straight:

- The decision is binary: accept or reject. We are not trying to estimate the true defective fraction of the lot.
- Sampling does not improve the quality of the lot. “You cannot inspect quality into the product.” The goal is to ensure the output as a whole conforms to requirements.
- Acceptance sampling is most useful when 100% inspection is impossible (destructive testing), expensive, slow, or itself error-prone (inspector fatigue across very long runs).

When to use. Inspection is necessary when process capability is not yet high enough to skip it, when the test is destructive, when the cost or time of 100% inspection is prohibitive, or when the supplier's quality history is excellent and a confirmatory check suffices.

Single, double, multiple, sequential. Sampling plans differ in how many stages of sampling they allow:

- *Single-sampling.* Inspect n items. Accept if defectives $d \leq c$, reject otherwise.
- *Double-sampling.* Inspect n_1 . If $d_1 \leq c_1$, accept. If $d_1 > c_2$, reject. Otherwise inspect a second sample of n_2 and accept if $d_1 + d_2 \leq c_2$.
- *Multiple-sampling.* Generalizes double-sampling to several stages.
- *Sequential-sampling.* Each individual item is inspected one at a time and contributes to a running statistic. The lot is accepted, rejected, or undecided after each item.

Plans with more stages can require less total inspection on average for the same protection (the OC curve is similar) but cost more in administrative complexity.

42 Single-Sampling Plan for Attributes

Single-Sampling Plan for Attributes

A single-sampling plan is specified by three numbers:

- N = lot size,
- n = sample size,
- c = acceptance number.

Inspect n items selected at random from the lot. Let d be the number of defective items observed. The decision rule is:

$$d \leq c \Rightarrow \text{accept the lot}, \quad d > c \Rightarrow \text{reject the lot}.$$

42.1 The Operating-Characteristic Curve

The probability that the plan accepts a lot whose true defective fraction is p is the *operating-characteristic* (OC) function $P_a(p)$. When N is much larger than n , the binomial approximation to the hypergeometric (Part II) gives

$$P_a(p) = P\{d \leq c\} = \sum_{d=0}^c \binom{n}{d} p^d (1-p)^{n-d}.$$

For smaller N the exact form uses the hypergeometric directly:

$$P_a(p) = \sum_{d=0}^c \frac{\binom{Np}{d} \binom{N-Np}{n-d}}{\binom{N}{n}}.$$

Effects of n and c on the OC curve.

- Increasing n (with c/n held roughly constant) makes the OC curve *steeper*: lots with even slightly elevated p are rejected much more reliably. This is the same $1/\sqrt{n}$ behavior that ran through Parts III and IV.
- Decreasing c shifts the OC curve to the *left*: smaller defective fractions are required for acceptance.
- The *ideal* OC curve would drop vertically at the threshold $p = \text{AQL}$, accepting every better-than-AQL lot with probability 1 and rejecting every worse-than-AQL lot with probability 1. Real plans approximate this with a sigmoid curve, and steeper is better.

R companion: [15-acceptance-sampling-oc.R](#)¹⁶

```
library(AcceptanceSampling)
proportion_defective <- seq(0, 0.08, length = 50)
n50c1 <- OC2c(n = 50, c = 1, pd = proportion_defective)
n100c2 <- OC2c(n = 100, c = 2, pd = proportion_defective)
plot(n50c1, type = "l"); lines(proportion_defective, n100c2@paccept, col = "red")
```

42.2 AQL, LTPD, and the Two Risks

Two pairs of complementary points anchor the OC curve in practice:

¹⁶<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/15-acceptance-sampling-oc.R>

AQL, LTPD, α , β

- **Acceptable Quality Level (AQL).** The poorest process average the consumer is willing to accept as a long-run process target. A good plan accepts AQL lots with high probability $1 - \alpha$.
- **Producer's risk α .** The probability that an AQL lot is rejected. Typically $\alpha \in [0.01, 0.10]$.
- **Lot Tolerance Percent Defective (LTPD).** The poorest individual lot the consumer is willing to accept (also called the *rejectable quality limit*, RQL). A good plan accepts LTPD lots with low probability β .
- **Consumer's risk β .** The probability that an LTPD lot is accepted. Typically $\beta = 0.10$.

A plan is designed by specifying two of (AQL, α , LTPD, β) and reading off (n, c) from a table or calculation.

42.3 Advantages and Disadvantages

Advantages over 100% inspection	Disadvantages over 100% inspection
Less inspection $\Rightarrow$ less expense and less product handling damage.	Risk of accepting bad lots (β).
Solution for destructive testing.	Risk of rejecting good lots (α).
Fewer personnel needed.	Less information generated about the lot.
Reduced inspector error from fatigue.	Planning and documentation overhead.
Motivates supplier quality improvement.	

43 Double-Sampling Plans

A double-sampling plan defers a marginal decision to a second sample.

Double-Sampling Plan

Specify (n_1, c_1, n_2, c_2) with $c_1 < c_2$. Inspect n_1 items and let d_1 be the defectives observed.

- If $d_1 \leq c_1$, **accept** the lot.
- If $d_1 > c_2$, **reject** the lot.
- If $c_1 < d_1 \leq c_2$, inspect n_2 more items. Let d_2 be the defectives in the second sample.
 - If $d_1 + d_2 \leq c_2$, accept. Otherwise reject.

The advantage is the *average sample number* (ASN): for very good or very bad lots the decision is reached after n_1 alone, and only marginal lots require the second stage. For the same OC

curve, double-sampling can reduce the expected inspection burden compared with a matched single-sampling plan.

44 MIL-STD-105E (Attributes)

MIL-STD-105E is a system of single-, double-, and multiple-sampling plans for attribute inspection, indexed by lot size and AQL. The selection procedure is:

1. **Choose an inspection level.** General levels are I, II (default), and III, with III calling for the largest samples and the highest discrimination. Special levels S-1 through S-4 are used when small samples are required for reasons unrelated to discrimination (e.g. destructive testing).
2. **Look up a sample-size code letter** from the lot size and inspection level.
3. **Look up** (n, c) from the code letter and AQL, in three regimes: *normal*, *tightened*, or *reduced*.
4. **Apply switching rules.** Inspection starts under the normal regime. If two of any five consecutive lots are rejected, switch to tightened. On five consecutive accepted lots, return to normal. Reduced inspection is permitted after a long stretch of conforming lots if production is steady.

The standard's structure forces a feedback loop: persistently bad supplier quality makes the inspection burden heavier, which incentivizes the supplier to improve.

R companion: [16-mil-std-105e.R](#)¹⁷

```
library(AQLSchemes)
plan_normal <- AASingle("Normal")
plan_tightened <- AASingle("Tightened")
plan_oc <- OC2c(plan_normal$n, plan_normal$c, pd = proportion_defective)
plot(plan_oc, type = "l")
```

45 MIL-STD-414 (Variables)

When the quality characteristic is a continuous measurement (length, weight, voltage, viscosity), variables-sampling plans use a fewer-samples-for-the-same-protection advantage of variables data. MIL-STD-414 (now ANSI/ASQ Z1.9) provides plans indexed similarly to 105E.

Two procedures.

- *k-method (Procedure 1).* Compute the standardized distance from the sample mean $\bar{X}$ to the specification limit, e.g. $(USL - \bar{X})/s$ for a one-sided upper limit. Accept if this distance exceeds a tabulated k .
- *M-method (Procedure 2).* Estimate the proportion of the lot beyond the specification limit using $\bar{X}$, s , and a normal model. Accept if the estimate is below a tabulated maximum allowable percent defective M .

¹⁷<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/16-mil-std-105e.R>

The k -method is simpler. The M -method is more flexible when both upper and lower specification limits are present.

Comparison with 105E. For the same AQL and lot size, MIL-STD-414 typically calls for substantially smaller sample sizes than MIL-STD-105E because variables data carries more information per observation than the binary defective/non-defective label. The price is the assumption of a (near-)normal distribution of the quality characteristic.

Practice 13: Selecting a 105E Plan

A supplier ships a component in lots of size $N = 3000$. The AQL has been negotiated at 1%. Use MIL-STD-105E to find the normal, tightened, and reduced single-sampling plans for this situation under *general inspection level II*. Repeat for general inspection level I. Compare the sample sizes across the two inspection levels and across the three regimes, and plot the corresponding OC curves on a single axis to visualize the differences in producer's risk and consumer's risk.

Practice 14: A Variables Plan from MIL-STD-414

A military inspector requires a variables sampling plan with $AQL = 1.5\%$ and lot size $N = 7000$ under inspection level II. Assume the lot or process variability is unknown and that there is a single specification limit.

- Derive the normal and tightened single-sampling plans using Procedure 1 (k -method) of MIL-STD-414.
- Derive the corresponding plans from MIL-STD-105E for the same lot size and AQL under level II. Compare the sample sizes and discuss why the variables plan typically requires fewer observations.

46 Lot Formation and Random Sampling

Story: Stella Cunliffe at Guinness

Stella Vivian Cunliffe (1917–2012) joined the Dublin brewers Arthur Guinness Son & Co. as a statistician in 1947 and worked there until 1970. She took on the role of head of the statistics department in 1955. She later became the first woman to serve as President of the Royal Statistical Society.

Among her contributions at Guinness was a quality-control redesign that has since become a textbook example of how the physical layout of an inspection workstation can bias acceptance-sampling data. “She redesigned the instructions for quality control workers who were tasked to either accept or reject handmade beer barrels. Before Cunliffe’s redesign, workers accepted

barrels by rolling them downhill and rejected barrels by pushing them uphill, the more difficult task; thus, workers were biased to accept barrels even if they were flawed. Cunliffe redesigned the quality control work station so that it was equally easy to reject or accept a barrel, eliminating the prior bias and saving Guinness money in the process.”

Cunliffe expanded on the broader lesson in her 1975 Presidential Address to the Royal Statistical Society, published in 1976 under the title “Interaction.” The point is that an acceptance-sampling plan can be mathematically well defined and still be defeated when the ergonomics of the inspection workstation systematically rewards one decision over the other.

Source: “Stella Cunliffe,” Wikipedia; Cunliffe (1976), “Interaction” (Presidential Address), JRSS Series A, 139(1), pp. 1–19.

The validity of any sampling plan rests on the validity of the random sample. Three guidelines:

- *Lots should be homogeneous.* Material from a single shift, a single batch of raw material, and a single piece of equipment, ideally.
- *Larger lots are preferred.* A larger N allows a smaller relative sample fraction n/N to achieve the same OC curve.
- *Sampling must be physically random.* Stratified or systematic sampling within a lot may be needed if the lot is laid out in a way (cubes, strata, layers) that simple random sampling cannot easily access.

Acceptance sampling decides the fate of an arriving lot. But every lot has a producer, and the producer’s process is running continuously. The next part shifts the perspective from receiving inspection to in-process surveillance: as units come off the line one by one or in small subgroups, how do we tell whether the process generating them is still in control?

Part IX

Statistical Process Control

Weeks 12–13 · What We Cover in This Part

This part covers the operating end of quality assurance: a process is running over time and we want to detect, quickly, the moment at which it drifts out of control. We distinguish chance causes from assignable causes and introduce the Shewhart control chart. We define the average run length and the average time to signal, work through sensitizing rules, and contrast Phase I (limits constructed from in-control data) with Phase II (live monitoring). We then develop variables control charts ($\bar{X}$, R , s) and attributes charts (p , np , c , u), and close with process capability analysis.

Textbook: Montgomery, Ch. 5–7, §§8.1–8.4, §8.6, and Ch. 9.

Story: Deming and the Quality Revolution

In 1980 the NBC television network aired a documentary, “If Japan Can, Why Can’t We?,” that tried to explain how Japanese cars, steel, and electronics had surpassed American industry in both quality and cost. The documentary’s answer was, in large part, the 80-year-old American statistician W. Edwards Deming.

Deming had been consulting on industrial quality since 1939, but American senior management “had no interest in the ‘technical’ details of quality control.” The opening came from MacArthur’s reconstruction of Japan. In 1947, MacArthur’s staff located Deming as an expert on statistical sampling and sent him to teach the Japanese “how we do it in America.” Ichiro Ishikawa, head of the Japanese Union of Scientists and Engineers, arranged seminars where the senior management of major Japanese companies actually attended. Deming told them that within five years they could produce products that “would dominate markets all over the world.” The Japanese beat his prediction by nearly two years.

Deming’s central message was about the nature of variation. Every production line is variable, and Fisher’s analysis of variance can split the variability into two sources: *special causes* (specific, identifiable) and *common* (or environmental) *causes* (the background hum of poorly maintained machines, inconsistent raw materials, and uncontrolled working conditions). Deming insisted that common causes are the larger source and that “only management can reduce or remove” them. Setting ever-tighter tolerances on the output while ignoring the system that produces it, as in the 1970s “zero defect” slogan and the 1980s total-quality-management fad, was to Deming nothing more than “empty words and exhortations.”

The lesson for what follows is that the Shewhart control chart, which this part develops, is a tool for thinking about variation in Deming’s sense: distinguish the system from the operator, treat each plotted point as evidence about the system, and act on the cause that explains it. What the customer wants, Deming insisted, “is not a perfect product but a *reliable* product.”

Source: Salsburg (2001), *The Lady Tasting Tea*, ch. 24, pp. 247–253.

A process is running. Some of its output variation is inherent and unavoidable. Some is caused by identifiable sources that can in principle be fixed. How do we tell the two apart in real time, and how do we monitor the process so that an out-of-control condition is detected quickly? The Shewhart control chart monitors production using a chart of measurements over time with control

limits drawn at a fixed multiple of the standard deviation, and rests on a distinction between *chance causes* (random, inherent, unavoidable) and *assignable causes* (specific, identifiable, removable).

Statistical process control is a different mode of statistical thinking from hypothesis testing. The null hypothesis is the same (the process is in control), but the test is repeated continuously, and the practitioner is looking for the first moment at which the data depart from the null. The chart's elegance is that the operator on the floor needs no statistical training to use it: a point outside the control limits triggers an investigation, and so does a sequence of points showing a trend. The statistics is built into the limits. Together with acceptance sampling, SPC defines what “quality assurance” means in the title of this course, and it closes the loop with hypothesis testing: each plotted point is a test, and the control chart is a sequence of such tests with a fixed Type I error rate per point.

47 Chance and Assignable Causes

Two kinds of variation. Every measurable quality characteristic varies. Shewhart's central distinction:

Chance causes	Assignable causes
Natural, inherent variation.	Operator error.
Statistical noise.	Machine error or wear.
Many small contributions, none decisive.	Material defects.
Cannot be removed without changing the process.	Measurement error.
	Poor designs or procedures.

In and Out of Control

A process operating with only chance causes of variation is said to be in *statistical control*. A process operating in the presence of assignable causes is said to be *out of control*.

What a control chart does and does not do. The chart only *detects* that an assignable cause is likely present. Identifying which cause and removing it requires human investigation. The control chart is a statistical procedure that triggers root-cause analysis. The analysis itself is engineering.

48 The Shewhart Control Chart

Let w be a sample statistic computed from each subgroup of measurements (the sample mean, the sample range, the fraction nonconforming, etc.). Let μ_w and σ_w denote its mean and standard deviation under in-control conditions.

General Shewhart Control Chart

The Shewhart chart plots successive values of w against time, with three horizontal reference lines:

$$UCL = \mu_w + L \sigma_w \quad (\text{upper control limit}),$$

$$CL = \mu_w \quad (\text{center line}),$$

$$LCL = \mu_w - L \sigma_w \quad (\text{lower control limit}).$$

The constant L is the distance of the control limits from the center line, expressed in units of σ_w . The standard choice is $L = 3$, the *three-sigma* control limits, which under approximate normality of w catch only about 0.27% of in-control points outside the limits.

Connection to hypothesis testing. Each plotted point is implicitly a test of H_0 : “the process is in control” against H_1 : “the process has shifted.” Three-sigma limits set the Type I error rate per point at $\alpha \approx 0.0027$ (assuming approximate normality). A point outside the limits rejects H_0 for that subgroup.

Choice of control limits. Wider control limits decrease α (fewer false alarms) but increase β (slower detection of true shifts). The trade-off mirrors the one we made for hypothesis tests in Part V.

Warning limits. Some practitioners add two-sigma *warning limits* inside the three-sigma *action limits*. A single point between the warning and action limits is not in itself an alarm, but two consecutive such points are.

49 Average Run Length and Average Time to Signal

ARL and ATS

Let p be the probability that any single plotted point falls outside the control limits. The *Average Run Length* (ARL) is the expected number of points plotted before the first out-of-control signal:

$$\text{ARL} = \frac{1}{p}.$$

If subgroups are sampled at intervals of h time units, the *Average Time to Signal* (ATS) is

$$\text{ATS} = \text{ARL} \times h.$$

For a three-sigma in-control chart, $p \approx 0.0027$, so $\text{ARL}_0 \approx 370$. With $h = 30$ minutes, $\text{ATS}_0 \approx 185$ hours. When the process actually shifts, the ARL drops sharply: the steeper the shift, the faster the detection.

50 Sensitizing Rules

A single point outside the control limits is the standard signal. But subtler patterns inside the limits also indicate that the in-control assumption has been violated. The *Western Electric rules*

(Table 5.1 of the Statistical Quality Control Handbook) flag ten patterns:

1	One or more points outside the three-sigma control limits.
2	Two of three consecutive points outside the two-sigma warning limits but still inside the control limits.
3	Four of five consecutive points beyond the one-sigma limits.
4	A run of eight consecutive points on one side of the center line.
5	Six points in a row steadily increasing or decreasing.
6	Fifteen points in a row in zone C (within one sigma of CL).
7	Fourteen points in a row alternating up and down.
8	Eight points in a row on both sides of CL with none in zone C.
9	Any unusual or non-random pattern in the data.
10	One or more points near a warning or control limit.

Caveat. Stacking rules raises the false-alarm rate. Each rule has its own per-point Type I error contribution, and using all ten can drop the in-control ARL from 370 to under 100. In practice, charts use a small subset (typically rules 1, 2, 5, 6) chosen to detect the kinds of shifts most likely in the application.

51 Phase I and Phase II

A control chart is built in two phases.

Phase I (chart construction). Initial samples are collected to estimate μ_w and σ_w . Trial control limits are computed, the data are plotted against the trial limits, and any out-of-control points are investigated. Points whose causes are identified and corrected are discarded, the limits are recomputed, and the procedure iterates until all retained points plot in control. The resulting limits are adopted for routine use.

Phase II (process monitoring). The limits established in Phase I are applied to incoming data in real time. Out-of-control signals trigger investigation and corrective action, and the limits themselves are periodically re-validated.

52 Variables Control Charts: $\bar{X}$, R , s

When the quality characteristic is continuous, two charts are usually run together: one for the location ($\bar{X}$) and one for the spread (R or s).

52.1 $\bar{X}$ and R Charts

Suppose subgroups of size n are collected, with m subgroups in Phase I. For each subgroup, compute $\bar{X}_i$ (the mean) and $R_i = X_{(n)} - X_{(1)}$ (the range). Let

$$\bar{\bar{X}} = \frac{1}{m} \sum_{i=1}^m \bar{X}_i, \quad \bar{R} = \frac{1}{m} \sum_{i=1}^m R_i.$$

$\bar{X}$ - R Control Limits

The trial control limits are

$$\begin{aligned}\bar{X} \text{ chart: } UCL &= \bar{\bar{X}} + A_2\bar{R}, & R \text{ chart: } UCL &= D_4\bar{R}, \\ CL &= \bar{\bar{X}}, & CL &= \bar{R}, \\ LCL &= \bar{\bar{X}} - A_2\bar{R}, & LCL &= D_3\bar{R},\end{aligned}$$

where A_2 , D_3 , D_4 are tabulated constants depending on the subgroup size n . The estimate of the in-control standard deviation is $\hat{\sigma} = \bar{R}/d_2$, again with d_2 from the table.

The constants (A_2, D_3, D_4, d_2) embed the distribution of the relative range under normality. Selected values (for common subgroup sizes):

n	A_2	D_3	D_4	d_2
2	1.880	0	3.267	1.128
3	1.023	0	2.574	1.693
4	0.729	0	2.282	2.059
5	0.577	0	2.114	2.326
6	0.483	0	2.004	2.534
7	0.419	0.076	1.924	2.704
10	0.308	0.223	1.777	3.078

52.2 $\bar{X}$ and s Charts

For larger subgroups (typically $n > 10$) or variable subgroup sizes, the sample standard deviation s is preferred over the range. Let

$$\bar{s} = \frac{1}{m} \sum_{i=1}^m s_i.$$

 $\bar{X}$ - s Control Limits

$$\begin{aligned}\bar{X} \text{ chart: } UCL &= \bar{\bar{X}} + A_3\bar{s}, & s \text{ chart: } UCL &= B_4\bar{s}, \\ CL &= \bar{\bar{X}}, & CL &= \bar{s}, \\ LCL &= \bar{\bar{X}} - A_3\bar{s}, & LCL &= B_3\bar{s}.\end{aligned}$$

The estimate of the standard deviation is $\hat{\sigma} = \bar{s}/c_4$.

52.3 R vs. s

- R charts are simpler to construct and equivalent to s charts for small n (up to about 10).
- s charts are more sensitive to small shifts and required when subgroup sizes vary.

*R companion: 17-xbar-s-charts.R*¹⁸

```
library(qicharts2)
qic(y = Days, x = Date, data = quality_df, chart = "xbar")
```

¹⁸<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/17-xbar-s-charts.R>


```
qic(y = Days, x = Date, data = quality_df, chart = "s")
```

53 Attributes Control Charts: p , np , c , u

When the quality characteristic is binary or a count, attributes charts apply.

Defective vs. defect. A *defective* unit fails to meet specifications. A *defect* (or *nonconformity*) is a single instance of failing to meet specifications. A defective unit may have multiple defects.

53.1 p Chart (Fraction Nonconforming)

Let D_i be the number of nonconforming units in a sample of n taken at time i . The sample fraction nonconforming is $\hat{p}_i = D_i/n$, and under in-control conditions $\hat{p}_i$ has mean p and variance $p(1-p)/n$.

p Chart

With $\bar{p} = (\sum_i D_i)/(mn)$ as the Phase I estimate of p , the trial limits are

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}, \quad CL = \bar{p}, \quad LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

(replacing LCL by 0 if the formula gives a negative value).

The np chart plots $D_i = n\hat{p}_i$ directly when the sample size is constant. The formulas are equivalent up to a factor of n .

53.2 c Chart (Number of Defects)

When the inspection unit is fixed (one square meter of fabric, one printed circuit board) and the count of defects x is naturally Poisson, the c chart applies. Let $\bar{c} = (\sum_i x_i)/m$.

c Chart

Under the Poisson assumption, $E[x] = \text{Var}[x] = c$, so

$$UCL = \bar{c} + 3\sqrt{\bar{c}}, \quad CL = \bar{c}, \quad LCL = \bar{c} - 3\sqrt{\bar{c}}.$$

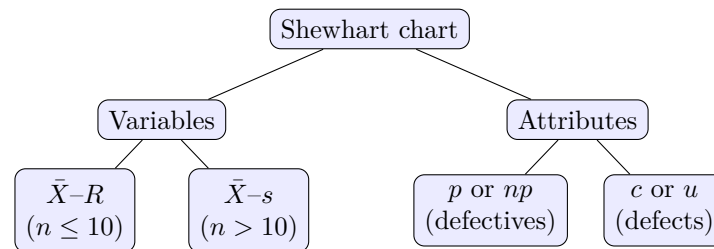
53.3 u Chart (Defects per Unit, Variable Inspection Size)

When the inspection unit varies (some samples cover more area, more time, more pages), normalize by the sample size n : $u_i = x_i/n$. Let $\bar{u} = (\sum_i u_i)/m$.

u Chart

$$UCL = \bar{u} + 3\sqrt{\bar{u}/n}, \quad CL = \bar{u}, \quad LCL = \bar{u} - 3\sqrt{\bar{u}/n}.$$

53.4 Decision Tree



Practice 15: Building Charts from Real Data

For each of the following situations, choose the appropriate Shewhart chart and justify your choice in two or three sentences.

- A call-center supervisor records the total number of incoming calls and the number of unanswered calls each day for 21 days. The supervisor wants to monitor the proportion of unanswered calls.
- A quality engineer at a wallpaper plant takes a 100-foot sample of wallpaper every hour and counts the number of printing defects (smears, distortions, missing ink) in the sample.
- A transcription company manager randomly selects 25 sets of pages from consecutive customer orders. Each set has a different number of pages, and the manager counts the number of typographical errors in each.

For each, write down the formulas for the trial control limits and describe how you would carry out the Phase I exercise.

*R companion: 18-attribute-charts.R*¹⁹

```

library(qicharts2)
qic(y = Num_Defective, n = Num_Items_in_Set, x = trial,
    data = bin_data_p_fixed, chart = "p")      # p chart, fixed n
qic(y = Num_of_Blemishes, n = Num_Units_Inspected, x = trial,
    data = u_data_var, chart = "u")           # u chart, variable n
qic(y = Counts, x = product_id, data = product_defect_counts, chart = "c")
  
```

54 Process Capability

A process can be in statistical control yet still produce too many out-of-specification units. *Capability* measures whether the in-control distribution fits inside the engineering specification limits.

Let USL and LSL be the upper and lower specification limits, μ the in-control process mean, and σ the in-control standard deviation.

¹⁹<https://github.com/aminrahimian/design-of-experiments-and-quality-assurance/blob/main/18-attribute-charts.R>

Process Capability Indices

$$C_p = \frac{USL - LSL}{6\sigma} \quad (\text{capability of a centered process}),$$

$$C_{pu} = \frac{USL - \mu}{3\sigma}, \quad C_{pl} = \frac{\mu - LSL}{3\sigma},$$

$$C_{pk} = \min(C_{pu}, C_{pl}) \quad (\text{realized capability of a possibly off-center process}).$$

$C_p \geq 1$ means the natural process spread fits inside the spec range. $C_p \geq 1.33$ is a common minimum for new processes, and $C_p \geq 2$ corresponds to “six sigma” capability.

C_p assumes the process is centered between LSL and USL , and C_{pk} does not. A process with $C_p = 2$ but μ shifted to the upper specification limit has $C_{pk} = 0$: the spread is small enough, but the centering is wrong.

Practice 16: Capability of a Golf-Ball Production Line

A quality engineer is auditing a golf-ball production line. The manager randomly draws three balls from each shift over a 50-day period and measures the diameter of each. The in-control mean diameter is approximately $\bar{\bar{X}} = 1.690$ inches with within-shift standard deviation $\hat{\sigma} = 0.001$ inches.

- (a) Compute C_{pk} if the specification limits are $USL = 1.695$, $LSL = 1.685$.
- (b) Repeat for $USL = 1.6925$, $LSL = 1.6875$.
- (c) Repeat for $USL = 1.7000$, $LSL = 1.6000$.
- (d) Discuss how the specification limits change the perceived capability of the process. Which scenario calls for the tightest process improvement effort?

References

- Primary textbook.** Douglas C. Montgomery. *Introduction to Statistical Quality Control*. 8th edition. Wiley, 2019. (Chapter and section pointers in each part's overview box refer to this edition.)
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- Quality control.** Acheson J. Duncan. *Quality Control and Industrial Statistics*. 5th edition. Irwin, 1986.